

Hemodynamics regulate spatiotemporal artery muscularization in the developing circle of Willis

Reviewed Preprint

v2 • June 25, 2024

Revised by authors

Reviewed Preprint

v1 • February 2, 2024

Siyuan Cheng, Ivan Fan Xia, Renate Wanner, Javier Abello, Amber N. Stratman, Stefania Nicoli 

Department of Genetics, Yale School of Medicine, 333 Cedar St, New Haven, CT 06520, USA • Yale Cardiovascular Research Center, Section of Cardiology, Department of Internal Medicine, Yale School of Medicine, 300 George St, New Haven, CT 06511, USA • Vascular Biology & Therapeutics Program, Yale School of Medicine, 10 Amistad St, New Haven, CT 06520, USA • Department of Cell Biology & Physiology, School of Medicine, Washington University in St. Louis, 660 S. Euclid Ave, St. Louis, MO 63110, USA

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

Abstract

Vascular smooth muscle cells (VSMCs) envelop vertebrate brain arteries, playing a crucial role in regulating cerebral blood flow and neurovascular coupling. The dedifferentiation of VSMCs is implicated in cerebrovascular diseases and neurodegeneration. Despite its importance, the process of VSMC differentiation on brain arteries during development remains inadequately characterized. Understanding this process could aid in reprogramming and regenerating differentiated VSMCs in cerebrovascular diseases. In this study, we investigated VSMC differentiation on the zebrafish circle of Willis (CoW), comprising major arteries that supply blood to the vertebrate brain. We observed that the arterial expression of CoW endothelial cells (ECs) occurs after their migration from the cranial venous plexus to form CoW arteries. Subsequently, *acta2*⁺ VSMCs differentiate from *pdgfrb*⁺ mural cell progenitors upon recruitment to CoW arteries. The progression of VSMC differentiation exhibits a spatiotemporal pattern, advancing from anterior to posterior CoW arteries. Analysis of blood flow suggests that earlier VSMC differentiation in anterior CoW arteries correlates with higher red blood cell velocity wall shear stress. Furthermore, pulsatile blood flow is required for differentiation of human brain *pdgfrb*⁺ mural cells into VSMCs as well as VSMC differentiation on zebrafish CoW arteries. Consistently, the flow-responsive transcription factor *klf2a* is activated in ECs of CoW arteries prior to VSMC differentiation, and *klf2a* knockdown delays VSMC differentiation on anterior CoW arteries. In summary, our findings highlight the role of blood flow activation of endothelial *klf2a* as a mechanism regulating the initial VSMC differentiation on vertebrate brain arteries.

eLife assessment

This study provides the first analysis of vascular stabilization on the critical and evolutionarily conserved structure around the Circle of Willis in the brain, strengthened by using parallel in vivo and in vitro experimental approaches. The evidence supporting the claims is **solid** and the work will be **valuable** for scientists studying developmental and disease-related vascular stabilization.

<https://doi.org/10.7554/eLife.94094.2.sa2>

Introduction

Vascular smooth muscle cells (VSMCs) are contractile mural cells wrapping around endothelial cells (ECs) of large vessels, especially arteries (Ando et al., 2022 [↗](#); Donadon & Santoro, 2021 [↗](#); Stratman et al., 2020 [↗](#)). The expression of contractile proteins, such as alpha-smooth muscle actin (encoded by *acta2*), distinguishes VSMCs from pericytes, the other type of mural cells primarily associated with small vessels (Bahrami & Childs, 2020 [↗](#); Donadon & Santoro, 2021 [↗](#); Stratman et al., 2020 [↗](#)). VSMCs are important for arterial homeostasis (Basatemur, Jorgensen, Clarke, Bennett, & Mallat, 2019 [↗](#)). In the brain, VSMC constriction and relaxation is essential for functional hyperemia in neurovascular coupling (Hill et al., 2015 [↗](#); Kaplan, Chow, & Gu, 2020 [↗](#)). Phenotype switching or dedifferentiation of VSMCs, marked by lower expression of contractile proteins, is prominent in pathological progression of atherosclerosis (Basatemur et al., 2019 [↗](#)). VSMC phenotype switching or dedifferentiation also contributes to cerebrovascular diseases and neurodegeneration (Aguilar-Pineda et al., 2021 [↗](#); Chou et al., 2022 [↗](#); Frosen & Joutel, 2018 [↗](#); Milewicz et al., 2010 [↗](#); Oka et al., 2020 [↗](#); Poitvein et al., 2014 [↗](#)). As development and regeneration share many common molecular and cellular mechanisms (Nowotarski & Sánchez Alvarado, 2016 [↗](#)), insights on VSMC differentiation on brain arteries during development may inform strategies to reprogram and regenerate dedifferentiated VSMCs and alleviate cerebrovascular diseases.

VSMCs differentiate from heterogeneous perivascular progenitors of mesoderm and neural crest origins (Ando et al., 2019 [↗](#); Donadon & Santoro, 2021 [↗](#); Whitesell et al., 2019 [↗](#)). Previous research described various mechanisms that regulate VSMC differentiation. Autonomous Notch activation is required for specification of *pdgfrb*⁺ mural cell progenitors from peri-arterial mesenchyme, and these progenitors later differentiate into *acta2*⁺ VSMCs (Ando et al., 2019 [↗](#)). Arterial Notch signaling activated by blood flow is necessary for *acta2*⁺ VSMC emergence on zebrafish trunk dorsal aorta (X. Chen, Gays, Milia, & Santoro, 2017 [↗](#)). Chemokine signaling promotes VSMC association with zebrafish dorsal aorta, whereas blood flow responsive transcription factor krüppel-like factor 2 (encoded by *klf2a* in zebrafish) prevents their association with the adjacent cardinal vein (Stratman et al., 2020 [↗](#)). Endothelial BMP signaling and autonomous *foxc1* expression regulate VSMC differentiation in zebrafish ventral head (Watterston, Zeng, Onabadejo, & Childs, 2019 [↗](#); Whitesell et al., 2019 [↗](#)). Autonomous expression of an ATP-sensitive potassium channel modulates VSMC differentiation from *pdgfrb*⁺ progenitors in the brain (Ando et al., 2022 [↗](#)). These studies suggest that VSMC differentiation is highly organotypic.

The circle of Willis (CoW) consists of major arteries that supply blood to the vertebrate brain, including internal carotid arteries (ICAs) and posterior communicating arteries (PCAs) (Campbell et al., 2019 [↗](#); Schröder, Moser, & Huggenberger, 2020 [↗](#)). In early development, ICAs are the first large arteries to form and supply all circulation in the developing brain (Kathuria, Gregg, Chen, & Gandhi, 2011 [↗](#); Menshawi, Mohr, & Gutierrez, 2015 [↗](#)). In adults, posterior circulation becomes

independent, and ICAs remain major feeding arteries in anterior, while connected to posterior circulation with PCAs, completing a circle of arteries (Kathuria et al., 2011; Schröder et al., 2020). In adults, intracranial aneurysms are frequently found close to CoW arteries (Frosen & Joutel, 2018). CoW arteries are wrapped by VSMCs, and VSMC dedifferentiation and hyperplasia are described in carotid atherosclerosis and Moyamoya disease (Chou et al., 2022; Fox, Dorschel, Lawton, & Wanebo, 2021). Spatiotemporal dynamics of VSMC differentiation on vertebrate CoW arteries during development have not been thoroughly investigated. VSMC differentiation is difficult to observe *in vivo* with mammalian models, as mammalian embryos develop in utero and depend on maternal circulation. Zebrafish is an important vertebrate model for vascular development, as zebrafish embryos are externally fertilized, optically clear, and thus accessible to confocal live imaging of developing blood vessels (Stratman & Weinstein, 2021). Establishment of zebrafish fluorescent transgenic reporter lines with *pdgfrb* and *acta2* promoters enables *in vivo* visualization and assessment of mural cells and VSMCs (Ando et al., 2016; Whitesell et al., 2014). In addition, zebrafish CoW shows comparable geometry to human CoW: the caudal division of the internal carotid artery (CaDI) in zebrafish is comparable to human ICA, and the posterior communicating segment (PCS) in zebrafish is comparable to human PCA (Isogai, Horiguchi, & Weinstein, 2001). Taking advantage of the zebrafish model, we describe the spatiotemporal pattern of VSMC differentiation on CoW arteries, and associate this pattern with blood flow activation of *klf2a* in arterial ECs.

Zebrafish COW vessel nomenclature	
CaDI	Caudal division of the internal carotid artery
BCA	Basal communicating artery
PCS	Posterior communicating segment

Results

Artery muscularization is spatiotemporally regulated in CoW arteries

In zebrafish brain, arteries are formed by ECs from cranial venous plexus (Fujita et al., 2011). Recent single-cell transcriptome profiling of vascular cells in prenatal human brain also supports an endothelial differentiation trajectory from proliferative venous ECs to quiescent arterial ECs (Crouch et al., 2022). To establish the temporal sequence of CoW arterial specification and VSMC differentiation, we started with confocal live imaging of fluorescent transgenic reporter lines *Tg(flt4:yfp)^{hu4881}*, which labels primitive venous ECs, and *Tg(kdr:l:hras-mcherry)^{s896}*, which is expressed in all ECs but enriched in arterial ECs, at four stages starting at 32 hours post fertilization (hpf), when CoW arteries are assembled (Chi et al., 2008; Hogan et al., 2009; Isogai et al., 2001). As arterial ECs express higher *kdr*, we used the intensity of *mcherry* as an indicator of artery specification (Figure 1A–1E) (Chi et al., 2008). We found that when ECs from cranial venous plexus migrated to form CoW arteries at 32 hpf, they still retained primitive venous identity, as indicated by *flt4* expression (Figure 1A). When connections between cranial venous plexus and CoW arteries disappeared at 54 hpf, CoW arteries no longer expressed primitive venous *flt4* (Figure 1B). Expression of arterial enriched *kdr* in CaDI and PCS at 3 days post fertilization (dpf) was significantly higher than 32 hpf (Figure 1C and 1E). At 4 dpf, *kdr* expression in CoW arteries remained comparable to 3 dpf, suggesting that arterial specification of CoW was completed by 3 dpf (Figure 1C–1E). Together, these data suggest CoW morphogenesis precedes arterial endothelial gene expression.

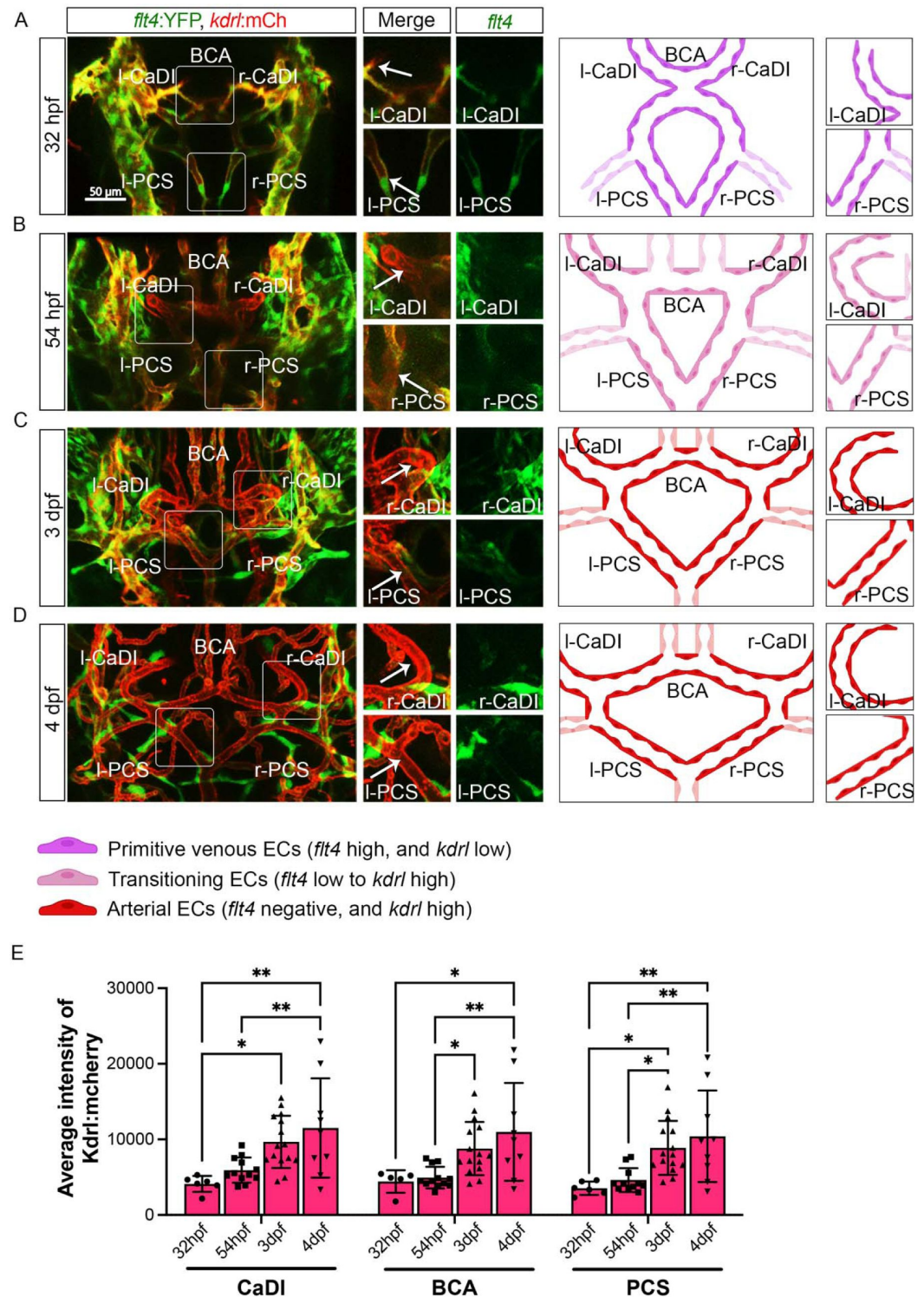


Figure 1.

Artery gene expression of endothelial cells (ECs) in circle of Willis (CoW) arteries

(A-D) Confocal live images of Tg(*flt4:yfp*, *kdrl:hras-mcherry*)^{hu4881/s896} and scheme representation of CoW arteries in zebrafish brain at 32 hour post fertilization (hpf) (A), 54 hpf (B), 3 day post fertilization (dpf) (C), and 4 dpf (D). Green channel represents *flt4:yfp* fluorescence, Red channel represents *kdrl:hras-mcherry* fluorescence, and Merge panel

combines both channels. Arrows point to the CoW arteries with *kdrl:hras-mcherry* signal. Scale bar = 50 μ m

(E) Average intensity of *kdrl:hras-mcherry* in caudal division of internal carotid arteries (CaDI), basal communicating artery (BCA), and posterior communicating segments (PCS) at 32 hpf (n=6, 2 independent experiments), 54 hpf (n=12, 4 independent experiments), 3 dpf (n=15, 4 independent experiments), and 4 dpf (n=9, 2 independent experiments), two-way analyses of variance followed by Tukey's multiple comparisons, represented with mean $\pm$ SD, * $p \leq 0.05$, ** $p \leq 0.01$

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, EC: endothelial cell, l-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, l-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

Previous research suggested that *acta2*⁺ VSMCs on CoW arteries differentiate from *pdgfrb*⁺ mural cell progenitors (Ando et al., 2022 [↗](#); Ando et al., 2019 [↗](#)). Thus, to characterize spatiotemporal dynamics of VSMC differentiation on CoW arteries, we performed confocal live imaging of Tg(*acta2:mcherry*)^{ca8}, TgBAC(*pdgfrb:egfp*)^{ncv22}, and Tg(*kdrl:cerulean*)^{sd24}, to visualize respectively VSMCs (red), *pdgfrb*⁺ progenitors (green), and developing arteries (white) at four stages starting from 32 hpf (Figure 2 [↗](#)) (Ando et al., 2016 [↗](#); Page et al., 2013 [↗](#); Whitesell et al., 2014 [↗](#)). We found that *pdgfrb*⁺ progenitors emerged on CoW arteries around 54 hpf after CoW morphogenesis, and most *pdgfrb*⁺ progenitors did not express *acta2* at 54 hpf (Figure 2A [↗](#)-2B, 2E-2F [↗](#), and S1A-S1B [↗](#)). The number of *pdgfrb*⁺ progenitors on CoW arteries increased temporally from 54 hpf to 3 dpf (Figure 2B [↗](#), 2C [↗](#), and 2E [↗](#)). Notably, while many *pdgfrb*⁺ cells on CaDI differentiated into *acta2*⁺ VSMCs by 3 dpf, *acta2*⁺ VSMCs were hardly observed on adjacent BCA or PCS, suggesting that VSMC differentiation started between 54 hpf and 3 dpf on CaDI but not BCA or PCS (Figure 2B [↗](#)-2C [↗](#), 2E [↗](#)-2F [↗](#), and S1A-S1B [↗](#)). At 4 dpf, the number of *pdgfrb*⁺ cells on CoW arteries remained comparable to 3 dpf, while the number of *acta2*⁺ VSMCs on CaDI increased (Figure 2C [↗](#)-2F [↗](#) and S1A-S1B [↗](#)). Importantly, most *pdgfrb*⁺ cells on BCA and PCS only started to express *acta2* around 4 dpf, much later than CaDI (Figure 2C [↗](#)-2F [↗](#) and S1A-S1B [↗](#)). These findings support that differentiation of *acta2*⁺ VSMCs from *pdgfrb*⁺ progenitors progresses spatially and temporally along the anterior to posterior direction in CoW arteries.

Figure 2.

Vascular smooth muscle cell (VSMC) differentiation on circle of Willis (CoW) arteries

(A-D) Confocal live images of Tg(*acta2:mcherry*, *kdr1:cerulean*)^{ca8/sd24} TgBAC(*pdgfrb:egfp*)^{ncv22} and scheme representation of vascular endothelium and mural cells on CoW arteries in zebrafish brain at 32 hour post fertilization (hpf) (A), 54 hpf (B), 3 day post fertilization (dpf) (C), and 4 dpf (D). White channel represents *kdr1:cerulean* fluorescence, Red channel represents *acta2:mcherry*, Green channel represents *pdgfrb:egfp*, Merge 1 panel combines all three channels, Merge 2 combines *acta2:mcherry* in red and *pdgfrb:egfp* in green. Arrows point to the CoW arteries with *pdgfrb:egfp* and *acta2:mcherry* signal. Scale bar = 50 μ m

(E) Number of *pdgfrb*⁺ vascular mural cell progenitors per 100 μ m vessel length on caudal division of internal carotid arteries (CaDI), basal communicating artery (BCA), and posterior communicating segments (PCS) at 32 hpf (n=3, 1 independent experiment), 54 hpf (n=6, 1 independent experiment), 3 dpf (n=6, 1 independent experiment), and 4 dpf (n=5, 1 independent experiment), two-way analyses of variance followed by Tukey's multiple comparisons, represented with mean $\pm$ SD, ** p $\leq$ 0.01, *** p $\leq$ 0.001, **** p $\leq$ 0.0001

(F) Number of *acta2*⁺ VSMCs per 100 μ m vessel length on CaDI, BCA, and PCS at 32 hpf (n=5, 1 independent experiment), 54 hpf (n=6, 2 independent experiments), 3 dpf (n=24, 6 independent experiments), and 4 dpf (n=25, 6 independent experiments), two-way analyses of variance followed by Tukey's multiple comparisons, represented with mean $\pm$ SD, **** p $\leq$ 0.0001

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, EC: endothelial cell, VSMC: vascular smooth muscle cell, l-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, l-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

CoW arteries have spatiotemporal difference in hemodynamics

Our data suggest that VSMCs on CaDI of anterior CoW differentiate much earlier than BCA and PCS of posterior CoW, although all *acta2*⁺ VSMCs on CoW differentiated from *pdgfrb*⁺ progenitors after they were recruited to CoW (Figure 2 and S1). In addition, anterior and posterior CoW vascular morphogenesis and arterial specification occurred around the same time (Figure 1). We speculate that hemodynamic distribution may contribute to the spatiotemporal difference in VSMC differentiation on CoW arteries. CaDI receive proximal arterial feed through lateral dorsal aorta from cardiac outflow tract (Isogai et al., 2001). Furthermore, CaDI differs from the BCA or PCS in terms of overall vascular geometry, which is among determinants of local vascular hemodynamic forces such as wall shear stress (WSS) (Katritsis et al., 2007). To explore this possibility, we performed microangiography with a high molecular weight fluorescent dextran to determine each CoW artery diameter. We found that there was no significant difference among CaDI, BCA and PCS (Figure 3A–3D). Then, we measure red blood cell (RBC) velocity by axial line scanning microscopy to analyze Tg(*kdr1:gfp*; *gata1:DsRed*)^{zn1/sd2} zebrafish at 54 hpf, 3 and 4 dpf. The RBC velocity across CoW arteries was differentially distributed, with CaDI having the highest RBC velocity from 54 hpf to 4 dpf (Figure 3E–3G). This is consistent with CaDI being the inlet vessel for arterial blood flow within the entire CoW circulation. Next, we used computational fluid dynamics (CFD) to simulate the effect of flow on CoW, based on the Dextran-filled CoW geometric network and RBC velocity (ANSYS, 2014; Z. Chen et al., 2019; Fernandes et al., 2022). We found an overall increase in average WSS from 54 hpf to 4 dpf (Figure 3A–3C, and 3H). Further, the simulated flow pattern and WSS on the CoW showed that WSS in CaDI was significantly higher than BCA and PCS at 54 hpf, 3 dpf, and 4 dpf (Figure 3A–3C, 3I–3K). Moreover, upon comparing WSS with the number of *acta2*⁺ VSMCs per CoW arteries during the development, we observed a Pearson correlation coefficient of $r = 0.595$, indicating a

moderate to strong positive correlation between these two variables (**Figure S1C**). In summary, the analysis of blood flow suggests that CoW arteries undergo varied hemodynamic WSS, which correlates with the spatial and temporal pattern of VSMC differentiation along CoW arteries.

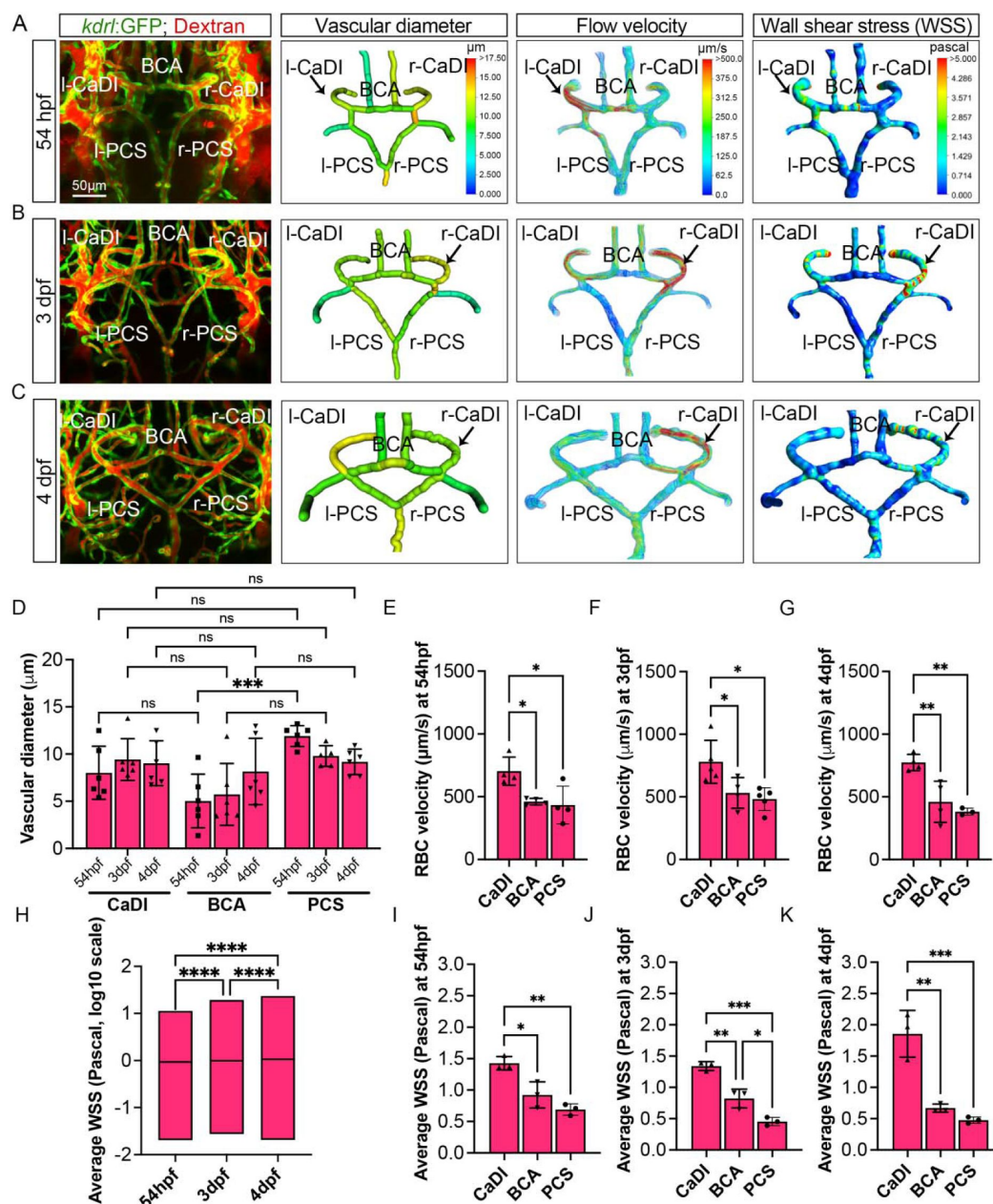


Figure 3.

Computational fluid dynamic (CFD) simulation of circle of Willis (CoW) arteries

(A-C) Confocal live images of *Tg(kdrl:gfp)^{zn1}* injected with dextran and representation of vascular diameter, CFD simulated flow and wall shear stress (WSS) within the CoW arteries at 54 hour post fertilization (hpf) (A), 3 day post fertilization (dpf) (B), and 4 dpf (C).

(C). Arrows point to the CoW arteries with high flow velocity and WSS. Scale bar = 50 μm

(D) Average vascular diameter in caudal division of internal carotid arteries (CaDI), basal communicating artery (BCA), and posterior communicating segments (PCS) at 54 hpf (n=6, 1 independent experiment), 3 dpf (n=6, 1 independent experiment), and 4 dpf (n=6, 1 independent experiment), two-way analyses of variance followed by Tukey's multiple comparisons, *** $p \leq 0.001$

(E-G) Red blood cell (RBC) velocity in CaDI, BCA, and PCS at 54 hpf (n=4, 1 independent experiment) (E), 3 dpf (n=5, 1 independent experiment) (F), and 4 dpf (n=4, 1 independent experiment) (G), ordinary one-way ANOVA analyses with Tukey's multiple comparisons, represented with mean $\pm$ SD, * $p \leq 0.05$, ** $p \leq 0.01$

(H) Average wall shear stress (WSS) throughout the CoW arteries at 54 hpf (n=3, 1 independent experiment), 3 dpf (n=3, 1 independent experiment), and 4 dpf (n=3, 1 independent experiment), ordinary one-way ANOVA analyses with Tukey's multiple comparisons, **** $p \leq 0.0001$

(I-K) Average wall shear stress (WSS) in CaDI, BCA, and PCS at 54 hpf (n=3, 1 independent experiment) (I), 3 dpf (n=3, 1 independent experiment) (J), and 4 dpf (n=3, 1 independent experiment) (K), ordinary one-way ANOVA analyses with Tukey's multiple comparisons, represented with mean $\pm$ SD, * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, RBC: red blood cell, WSS: wall shear stress, l-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, l-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

Blood flow is required for CoW artery muscularization

Our data suggest that ECs under arterial flow might favor *acta2*⁺ VSMC differentiation from *pdgfrb*⁺ progenitors. To test this hypothesis, we set up an *in vitro* cell co-culture experiment where GFP-PDGFRB⁺ human brain pericytes (GFP-HBVPs) were cultured in flow amenable outlet slides, then covered by a thin layer of collagen type 1. Confluent human endothelial cells (HUVECs) were seeded on top of that and then exposed to steady-state laminar or pulsatile flow conditions (**Figure 4A** [↗](#)) (Abello, Raghavan, Yien, & Stratman, 2022 [↗](#)). Differentiation of PDGFRB⁺ human brain pericytes into ACTA2⁺ VSMCs was then quantified after 24 hours of flow treatment. We found that brain pericytes under pulsatile flow were larger and showed higher expression of VSMC differentiation markers including PDGFRB, ACTA2, and TRANSGLUTININ (Robin et al., 2013 [↗](#)) (**Figure 4B** [↗](#)–**4D** [↗](#)), suggesting that pulsatile flow favored PDGFRB⁺ progenitor differentiation into VSMC.

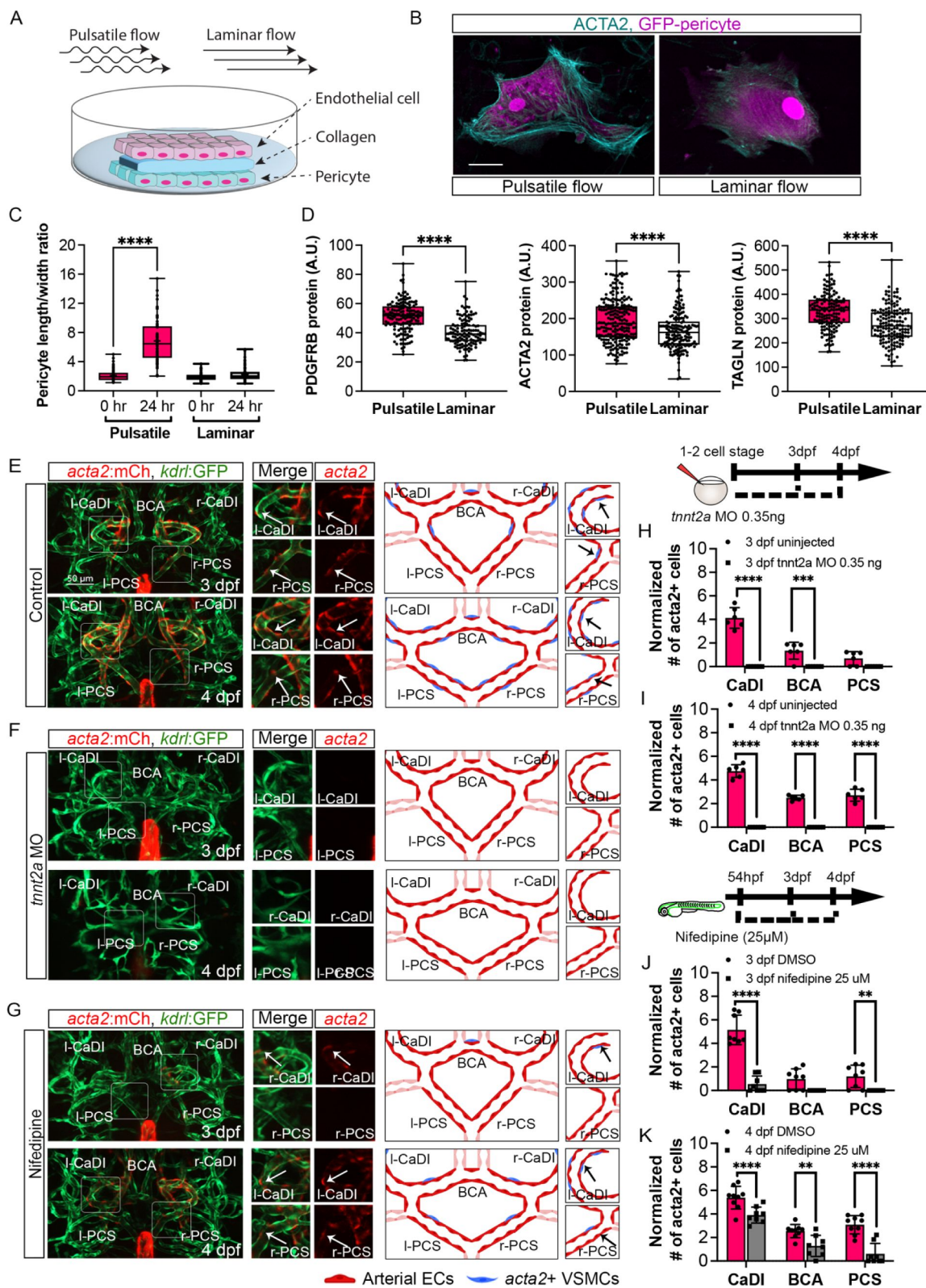


Figure 4.

Blood flow is required for vascular smooth muscle cell (VSMC) differentiation on circle of Willis (CoW) arteries

(A) Scheme representation of *in vitro* cell co-culture experiment

(B) Representative immunofluorescence images of brain pericytes after exposure of pulsative flow and laminar flow. Cells were stained for ACTA2 (cyan), and cytosolic GFP label (magenta). Scale bar = 10 μ m

(C) Morphological measurement of brain pericyte length/width ratio before and after exposure of pulsative flow and laminar flow (n = 63 cells), two-tailed Mann–Whitney test, represented with mean $\pm$ SD, **** p $\leq$ 0.0001

(D) Protein level of PDGFRB (n = 153 cells), ACTA2 (n = 222 cells), and TAGLN (n = 150 cells) in arbitrary unit (A.U.) after exposure of pulsative flow and laminar flow, two-tailed Mann–Whitney test, represented with mean $\pm$ SD, **** p $\leq$ 0.0001

(E–G) Confocal live images of Tg(*acta2:mcherry*; *kdr:lgfp*)^{ca8/zn1} and scheme representation of vascular endothelium and VSMCs on CoW arteries at 3 day post fertilization (dpf) and 4 dpf in control embryos (E), embryos injected with 0.35 ng *tnnt2a* morpholino (MO) (F), and embryos treated with 25 μ M nifedipine from 54 hpf (G). Red channel represents *acta2:mcherry*, Green channel represents *kdr:lgfp*, and Merge panel combines both channels. Arrows point to the CoW arteries with *acta2:mcherry* signal. Scale bar = 50 μ m

(H) Number of *acta2*+ VSMCs per 100 μ m vessel length on caudal division of internal carotid arteries (CaDI), basal communicating artery (BCA), and posterior communicating segments (PCS) at 3 dpf in uninjected control (n=6, 1 independent experiment) and embryos injected with 0.35 ng *tnnt2a* MO at one to two cell stage (n=6, 1 independent experiment), two-tailed Mann–Whitney test on each vessel's comparison, represented with mean $\pm$ SD, *** p $\leq$ 0.001, **** p $\leq$ 0.0001

(I) Number of *acta2*+ VSMCs per 100 μ m vessel length on CaDI, BCA, and PCS at 4 dpf in uninjected control (n=6, 1 independent experiment) and embryos injected with 0.35 ng *tnnt2a* MO at one to two cell stage (n=6, 1 independent experiment), two-tailed Mann–Whitney test on each vessel's comparison, represented with mean $\pm$ SD, **** p $\leq$ 0.0001

(I) Number of *acta2*+ VSMCs per 100 μ m vessel length on CaDI, BCA, and PCS at 3 dpf in DMSO control (n=8, 2 independent experiments) and embryos treated with 25 μ M nifedipine from 54 hpf (n=8, 2 independent experiments), two-tailed Mann–Whitney test on each vessel's comparison, represented with mean $\pm$ SD, ** p $\leq$ 0.01, **** p $\leq$ 0.0001

(J) Number of *acta2*+ VSMCs per 100 μ m vessel length on CaDI, BCA, and PCS at 4 dpf in DMSO control (n=9, 2 independent experiments) and embryos treated with 25 μ M nifedipine from 54 hpf (n=8, 2 independent experiments), two-tailed Mann–Whitney test on each vessel's comparison, represented with mean $\pm$ SD, ** p $\leq$ 0.01, **** p $\leq$ 0.0001

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, EC: endothelial cell, VSMC: vascular smooth muscle cell, MO: morpholino, l-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, l-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

Previous research suggests that blood flow is dispensable for recruitment of *pdgfrb*+ progenitors on BCA and PCS (Ando et al., 2016 [DOI](#)). To corroborate this, we injected 0.35 ng of *tnnt2a* morpholino antisense oligonucleotide (MO) into Tg(*pdgfrb:egfp*, *kdr:hras-mcherry*)^{ncv22/s896} at one

to two-cell stage, to abrogate cardiac contractility and thus blood flow (Ando et al., 2016 [↗](#); Chi et al., 2008 [↗](#); Sehnert et al., 2002 [↗](#)). Indeed, the number of *pdgfrb*⁺ cells on CaDI, BCA, and PCS at 3 dpf in *tnnt2a* morphants was comparable to uninjected sibling control, suggesting blood flow did not affect *pdgfrb*⁺ progenitor recruitment to CoW arteries (Figure S2G-S2H [↗](#)). To determine the effect of blood flow on VSMC differentiation, we injected *tnnt2a* MO into Tg(*acta2:mcherry*, *kdr1:gfp*)^{ca8/zn1} (Cross, Cook, Lin, Chen, & Rubinstein, 2003 [↗](#); Sehnert et al., 2002 [↗](#); Whitesell et al., 2014 [↗](#)). Notably, we found no *acta2*⁺ VSMCs on CoW arteries of *tnnt2a* morphants at either 3 or 4 dpf (Figure 4E [↗](#)-4F [↗](#) and 4H [↗](#)-4I [↗](#)), suggesting that blood flow was required for *acta2*⁺ VSMC differentiation on CoW arteries.

To further explore the temporal requirement of blood flow, we treated embryos with 25 μ M nifedipine, a drug shown to reduce heart rate in zebrafish embryos (Figure S2A-S2C [↗](#)) (Gierten et al., 2020 [↗](#)), from 54 hpf, right before the onset of *pdgfrb*⁺ progenitors' expression of *acta2* (Figure 2A [↗](#)-2C [↗](#)). We found that at 3 dpf the number of *acta2*⁺ VSMCs on the CaDI of treated embryos is greatly reduced (Figure 4G [↗](#) and 4J [↗](#)); *acta2*⁺ VSMCs on BCA and PCS at 4 dpf are also reduced significantly (Figure 4G [↗](#) and 4K [↗](#)), suggesting that blood flow reduction after 54 hpf impaired VSMC differentiation on CoW arteries.

To determine whether blood flow is required for short-term VSMC maintenance after differentiation on CoW arteries, we treated embryos with nifedipine from 4 dpf, after VSMC differentiation on CoW arteries, and imaged at 5 dpf. The number of *acta2*⁺ VSMCs on the CaDI and PCS in treated embryos was comparable to untreated controls (Figure S2D-S2F [↗](#)), suggesting that blood flow was required for differentiation but not for short-term maintenance of CoW VSMCs.

Blood flow-regulated transcription factor

klf2a modulates CoW artery muscularization

Our results suggest that CoW ECs might express a flow dependent transcriptional program that regulates spatiotemporal dynamics of VSMC differentiation on CoW arteries. Previous studies showed that expression pattern of transcription factor *Klf2* in arterial ECs closely follow the pattern of flow induced WSS (Lee et al., 2006 [↗](#); Parmar et al., 2006 [↗](#)). Consistently, higher *Klf2* expression can be induced by unidirectional pulsatile flow (Dekker et al., 2002 [↗](#)). Furthermore, *klf2* is implicated in VSMC migration on arterial blood vessels during mouse development: loss of *klf2* leads to aorta VSMC deficiency (Wu, Bohanan, Neumann, & Lingrel, 2008 [↗](#)). Interestingly, *klf2a* expression in primitive veins of zebrafish trunk prevents VSMC association (Stratman et al., 2020 [↗](#)). Hence, we tested the hypothesis that *klf2a* might be spatiotemporally regulated in CoW ECs and modulate VSMC differentiation. We first imaged Tg(*klf2a:h2b-egfp*, *kdr1:hras-mcherry*)^{ig11/s896}, which labels nuclei with active *klf2a* signaling, and quantified the number of *klf2a*⁺ ECs in CoW arteries (Chi et al., 2008 [↗](#); Steed et al., 2016 [↗](#)). To account for overall changes in EC number during development (Ulrich, Ma, Baker, & Torres-Vazquez, 2011 [↗](#)), and length of each vessel, we also imaged Tg(*fli1:nls-gfp*, *kdr1:hras-mcherry*)^{y7/s896}, which labels all EC nuclei (Figure S3A-S3I [↗](#)) (Roman et al., 2002 [↗](#)). From 32 hpf to 54 hpf, the number of *klf2a*⁺ EC in CaDI was comparable to PCS (Figure 5A [↗](#)-5C [↗](#)), however, the number of *klf2a*⁺ ECs in CaDI was significantly higher than BCA and PCS from 3 dpf (Figure 5A [↗](#) and 5D [↗](#)-5E [↗](#)), when *acta2*⁺ VSMCs differentiated on CaDI but not BCA and PCS (Figure 2B [↗](#)-2C [↗](#), 2F [↗](#), and S1A-S1C [↗](#)). Pearson correlation comparing the number of *klf2a*⁺ ECs with the number of *acta2*⁺ VSMCs in CaDI, BCA and PCS at 3 dpf showed strong positive correlation with $r = 0.974$ (Figure S5J). These results suggest that the number of *klf2a*⁺ ECs in CoW arteries is spatially correlated with VSMC differentiation.

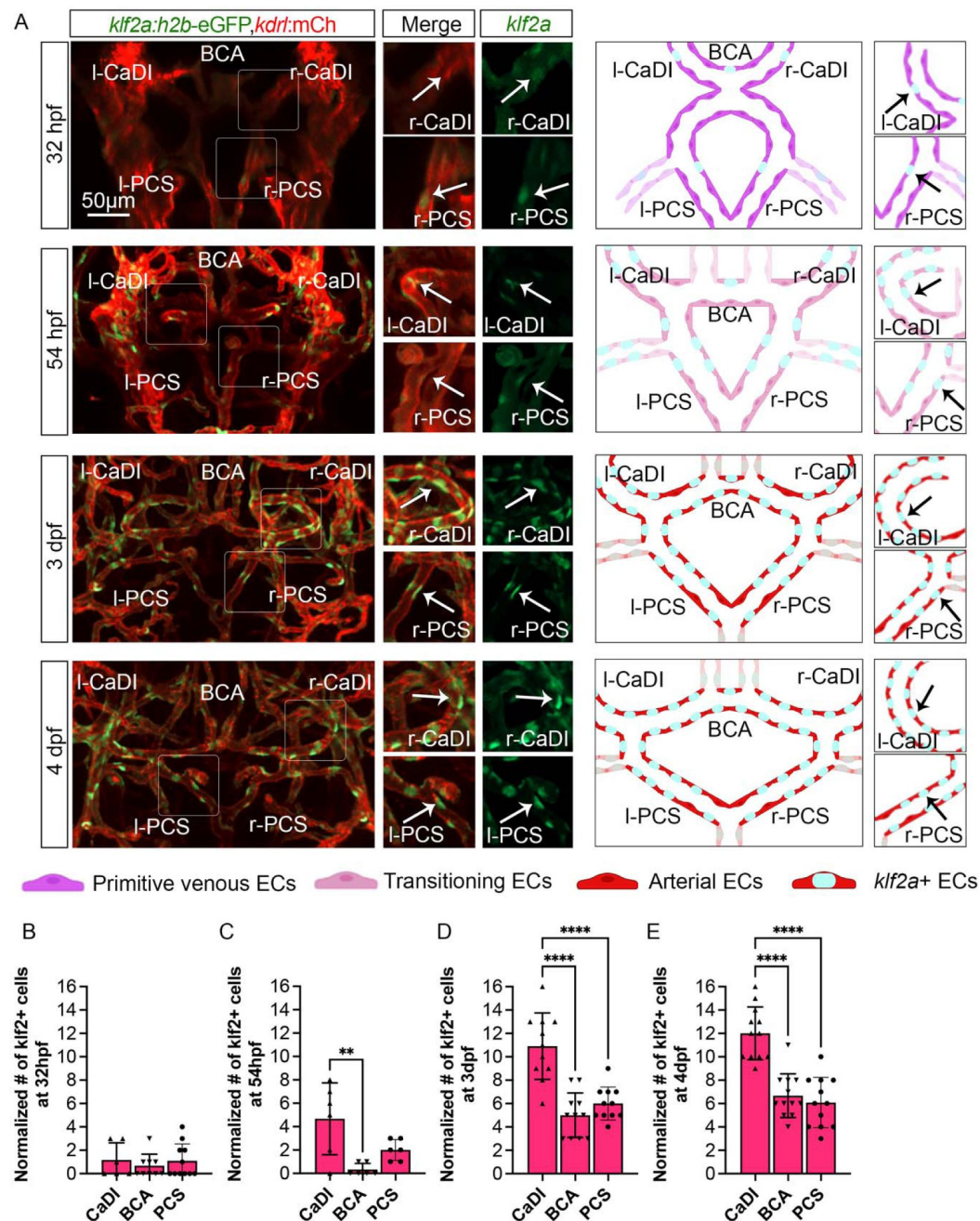


Figure 5.

Blood flow regulated transcription factor *klf2a* is expressed in circle of Willis (CoW) arteries

(A) Confocal live images of *Tg(klf2a:h2b-gfp, kdrl:ras-mcherry)^{ig11/s896}* and scheme representation of endothelial cells (ECs) in CoW arteries in zebrafish brain at 32 hour post fertilization (hpf), 54 hpf, 3 day post fertilization (dpf), and 4 dpf. Green channel represents *klf2a:h2b-gfp*, Red channel represents *kdrl:ras-mcherry*, and Merge panel combines both channels. Arrows point to the CoW arteries with *klf2a:h2b-gfp* signal. Scale bar = 50 μ m

(B-E) Number of *klf2a*+ ECs per 100 μ m vessel length on caudal division of internal carotid arteries (CaDI), basal communicating artery (BCA), and posterior communicating segments (PCS) at 32 hpf (n=6, 3 independent experiments) (B), 54 hpf (n=6, 2 independent experiments) (C), 3 dpf (n=11, 3 independent experiments) (D), and 4

dpf (n=12, 3 independent experiments) (E), ordinary one-way ANOVA analyses with Tukey's multiple comparisons, represented with mean $\pm$ SD, ** $p \leq 0.01$, **** $p \leq 0.0001$

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, EC: endothelial cell, l-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, l-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

To further define the role of *klf2a* in VSMC differentiation on CoW arteries, we knocked down *klf2a* in Tg(*acta2:mcherry*, *kdr:l:egfp*)^{ca8/zn1} with 11 ng MO injection at one to two-cell stage and imaged at 3 to 4 dpf (Nicoli et al., 2010 [\[link\]](#); Whitesell et al., 2014 [\[link\]](#); Zhong et al., 2006 [\[link\]](#)). We validated the effect of *klf2a* MO knockdown with Tg(*klf2a:h2b-egfp*, *kdr:l:hras-mcherry*)^{ig11/s896} (Figure S3G-S3I [\[link\]](#)) (Chi et al., 2008 [\[link\]](#); Steed et al., 2016 [\[link\]](#)). Compared to uninjected control, *klf2a* morphants have significantly less *acta2*⁺ VSMCs on CaDI at 3 dpf but a normal number by 4 dpf (Figure 6A [\[link\]](#)-6D [\[link\]](#)), suggesting that *klf2a* knockdown delayed VSMC differentiation on CaDI. Together these data suggest that endothelial *klf2a* is required for the onset of VSMC differentiation on CoW arteries.

Figure 6.

***klf2a* promotes vascular smooth muscle cell (VSMC) differentiation on anterior circle of Willis (CoW) arteries**

(A-B) Confocal live images of Tg(*acta2:mcherry*; *kdrl:gfp*)^{ca8/zn1} and scheme representation of vascular endothelium and VSMCs on CoW arteries at 3 day post fertilization (dpf) (A) and 4 dpf (B) in uninjected control embryos, embryos injected with 11 ng *klf2a* morpholino (MO). Red channel represents *acta2:mcherry*, Green channel represents *kdrl:gfp*, and Merge panel combines both channels. Arrows point to the CoW arteries with *acta2:mcherry* signal. Scale bar = 50 μ m

(C) Number of *acta2*+ VSMCs per 100 μ m vessel length on caudal division of internal carotid arteries (CaDI), basal communicating artery (BCA), and posterior communicating segments (PCS) at 3 dpf in uninjected control (n=8, 2 independent experiments) and embryos injected with 11 ng *klf2a* MO at one to two cell stage (n=8, 2 independent experiments), two-tailed Mann-Whitney test on each vessel's comparison, represented with mean $\pm$ SD, ** p $\leq$ 0.01

(D) Number of *acta2*+ VSMCs per 100 μ m vessel length on CaDI, BCA, and PCS at 4 dpf in uninjected control (n=8, 3 independent experiments) and embryos injected with 11 ng *klf2a* MO at one to two cell stage (n=8, 3 independent experiments), two-tailed Mann-Whitney test on each vessel's comparison, represented with mean $\pm$ SD

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, EC: endothelial cell, l-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, l-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

Discussion

In this study, we employed confocal live imaging of fluorescence transgenic zebrafish embryos to investigate the spatiotemporal dynamics of vascular smooth muscle cell (VSMC) differentiation on the cerebral vasculature (CoW), which comprises major arteries supplying blood to the vertebrate brain. Our observations revealed that CoW morphogenesis precedes the expression of arterial endothelial genes. Specifically, mural cell progenitors marked by *pdgfrb*+ initiate the expression of the VSMC marker *acta2* subsequent to their recruitment to CoW arteries. Notably, VSMC differentiation occurs earlier in the anterior CoW arteries compared to their posterior counterparts, owing to the elevated wall shear stress (WSS) resulting from a higher velocity of red blood cells in the incoming blood flow. To investigate the regulatory role of blood flow on the spatiotemporal dynamics of VSMC differentiation, we employed an in vitro co-culture assay along with genetic manipulation and drug treatments. Our findings indicate that blood flow indeed governs the timing and location of VSMC differentiation on CoW arteries. Moreover, we observed that the flow-responsive transcription factor *klf2a* is activated in a gradient manner from anterior to posterior CoW arteries, preceding the onset of VSMC differentiation. Knockdown experiments targeting *klf2a* revealed a delay in VSMC differentiation specifically in the anterior CoW arteries. Taken together, these findings highlight the role of endothelial *klf2a* activation by blood flow as a mechanism that promotes VSMC differentiation on CoW arteries within the brain (see [Figure 7](#) for a visual summary).

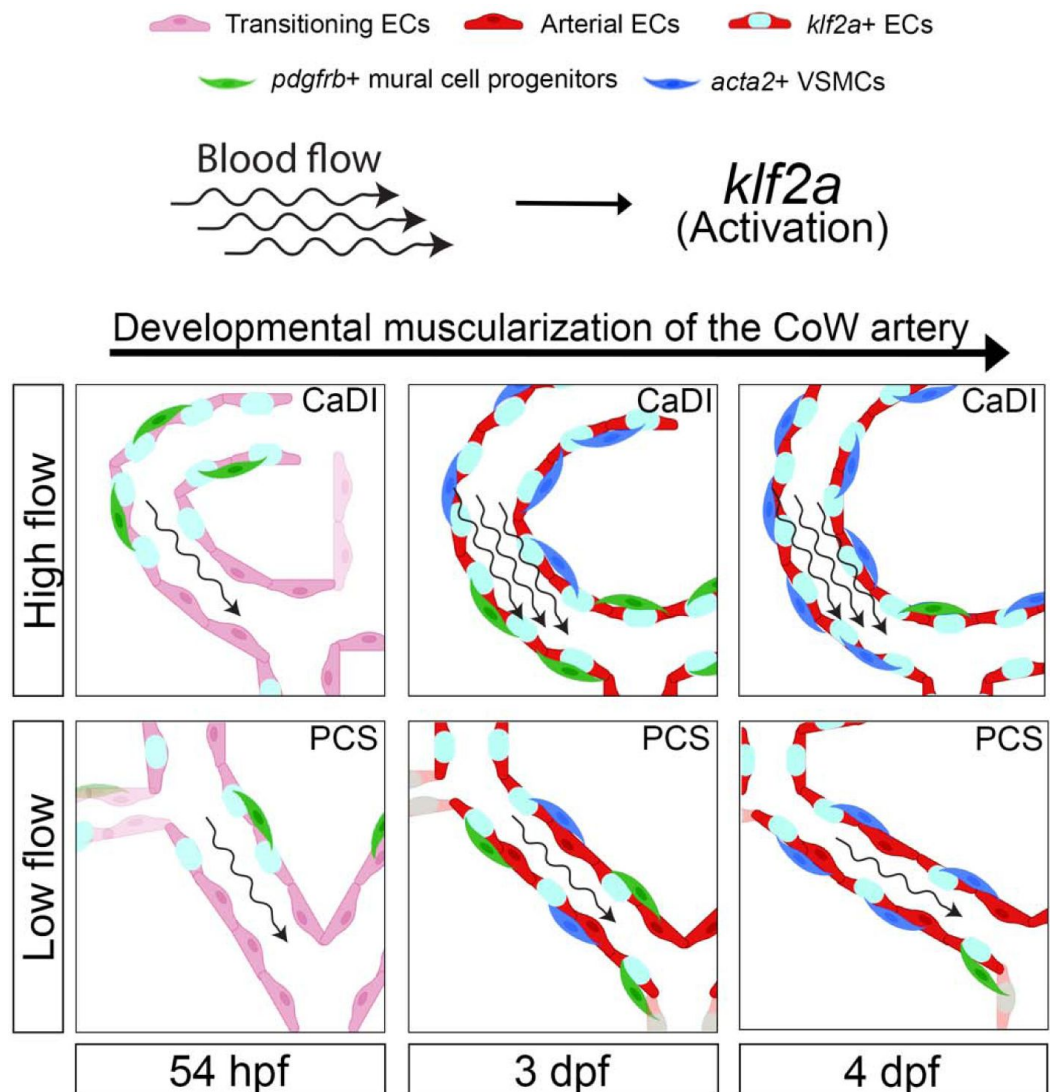


Figure 7.

Schematic model of the developmental muscularization of the CoW arteries

The model shows how blood flow generate higher hemodynamics in anterior CoW arteries like the caudal division of internal carotid artery (CaDI) via the activation of endothelial *klf2a* signaling. Other posterior CoW arteries with straight shape like the posterior communicating segment (PCS) experience less hemodynamic force and showed moderate *klf2a* activation and VSMC differentiation.

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, EC: endothelial cell, VSMC: vascular smooth muscle cell, CaDI: caudal division of internal carotid artery, PCS: posterior communicating segment

Previous research shows that in zebrafish trunk, as soon as recruited to the dorsal aorta, VSMCs express *acta2* and *transgelin* (*tagln*), suggesting simultaneous recruitment and differentiation from sclerotome progenitors (Ando et al., 2016; Stratman et al., 2017). On CoW arteries in zebrafish

brain, however, VSMC differentiation occurs after *pdgfrb*⁺ progenitor recruitment and proceeds from anterior to posterior (**Figure 2B** [↗](#)-**2F** [↗](#)). The observed spatiotemporal dynamics of CoW VSMC differentiation again highlights organotypic development of blood vessels.

Our data suggests *klf2a* mediate blood flow regulation of VSMC differentiation on brain arteries, and thus raises the question of how *klf2a* transduces endothelial signals to mural cell progenitors and VSMCs. Notch signaling appears a plausible downstream effector of *klf2a* activation, as previous research suggests Notch responds to flow in heart valve development (Fontana et al., 2020 [↗](#)), possibly downstream of *klf2* (Duchemin, Vignes, & Vermot, 2019 [↗](#)). Wnt signaling is another possible downstream effector of *klf2a*, as previous research suggests endocardial *klf2* upregulates Wnt signaling in neighboring mesenchymal cells in heart valve development (Goddard et al., 2017 [↗](#)). The roles of Notch and Wnt signaling in organotypic VSMC differentiation on brain arteries remain to be determined.

The expression of *klf2a* in CoW arterial ECs remained stable from 3 dpf to 4 dpf (**Figure 5** [↗](#)), which raises an interesting question on whether sustained *klf2a* expression supports further maturation of VSMCs, as *acta2*⁺ VSMCs on CaDI showed distinct morphology at 4 dpf compared with 3 dpf, when they started differentiation from *pdgfrb*⁺ mural cell progenitors (**Figure 2C** [↗](#)-**2D** [↗](#) and **S1** [↗](#)). Previous research found that *acta2*⁺ VSMCs on the BCA and PCS express pericyte enriched *abcc9* (ATP binding cassette subfamily C member 9) at 4 dpf (Ando et al., 2022 [↗](#)), when they started differentiation from *pdgfrb*⁺ mural cell progenitors and expressing *acta2*; *abcc9* expression is gradually lost from 5 dpf to 6 dpf (Ando et al., 2022 [↗](#)). In addition, *acta2*⁺ VSMCs on CaDI, BCA, and PCS still retain expression of *pdgfrb* at 6 dpf (Ando et al., 2021 [↗](#)). Thus, it is possible that stable expression of *klf2a* in CoW arterial ECs supports further VSMC maturation, as indicated by expression of *tagln* (Colijn, Nambara, & Stratman, 2023 [↗](#)). Another aspect of VSMC maturation is its function to confer vascular tone. A previous study found that VSMC covered vessels in zebrafish brain dilate as early as 4 dpf and constrict at 6 dpf (Bahrami & Childs, 2020 [↗](#)). Future study may focus on the association between expression of different VSMC markers and VSMC functional maturation.

Another question is how flow activates *klf2a* in brain arteries. In a hypoxia-induced pulmonary hypertension model, *klf2* is activated by G-protein coupled receptor mediated Apelin signaling (Chandra et al., 2011 [↗](#)). In heart valve development, endocardial *klf2a* is upregulated by membrane bound mechanosensitive channels *trpp2* and *trpv4* (Heckel et al., 2015 [↗](#)). The pathway through which flow activates *klf2a* in brain arterial ECs remains unknown. The CoW VSMC phenotype of *klf2a* morphants is similar to, but do not fully recapitulate the phenotypes of *tnnt2a* morphants or nifedipine treated embryos (**Figure 4E** [↗](#)-**4L** [↗](#) and **Figure 6** [↗](#)). A proximal explanation is compensation by paralogous *klf2b* in zebrafish. Further characterization of CoW VSMC development in *klf2a* and *klf2b* genetic mutants (Rasouli et al., 2018 [↗](#); Steed et al., 2016 [↗](#)) may help determine whether *klf2b* compensates *klf2a* in CoW VSMC differentiation.

In addition to WSS, transmural pressure and associated mechanical stretch is another mechanical input of vascular wall that may contribute to VSMC differentiation. VSMCs autonomously sense and adapt to mechanical stretch (Haga, Li, & Chien, 2007 [↗](#)). Cyclic stretching activates arterial VSMC production of certain components of extracellular matrix (ECM) (Leung, Glagov, & Mathews, 1976 [↗](#)). In turn, specific interactions between ECM and integrins enable VSMCs to sense mechanical stretch (Goldschmidt, McLeod, & Taylor, 2001 [↗](#); Wilson, Sudhir, & Ives, 1995 [↗](#)). Connection to ECM is essential for VSMC force generation (Milewicz et al., 2017 [↗](#)). VSMCs express stretch activated *trpp2*, which enables myogenic response to autoregulate resting arterial diameter (Sharif-Naeini et al., 2009 [↗](#)). Interestingly, ex vivo stretching promotes expression of contractile proteins, such as *acta2* and *tagln* in VSMCs on murine portal veins (Albinsson, Nordstrom, & Hellstrand, 2004 [↗](#); Turczynska, Hellstrand, Sward, & Albinsson, 2013 [↗](#)). In addition, excessive mechanical stretch promotes VSMC dedifferentiation and inflammation (Cao et al., 2017 [↗](#); Wang

et al., 2018 [\[1\]](#)). Further investigation is needed to determine the potential role of increasing transmural pressure and associated mechanical stretch during development in VSMC differentiation.

There are a few limitations to our current study. Genetic manipulation and drug treatment that reduce heart rate would also reduce nutrients carried by blood flow, and the effects of nutrient and flow reduction could not be uncoupled in live zebrafish embryos. In addition, these methods are only capable of qualitative reduction of flow but not specific dampening of pulsations. *In vitro* three-dimensional (3D) vascular culture models, which combine ECs and mural cells (Mirabella et al., 2017 [\[2\]](#); Vila Cuenca et al., 2021 [\[3\]](#)), could be further optimized to simulate complex geometry of brain arteries. Combining these models with microfluidics, which allows precise calibration of nutrient composition in culture media, flow velocity and pulse, would enable more thorough analysis of endothelial mechanotransduction and its contribution to VSMC differentiation (Abello, Raghavan, Yien, & Stratman, 2022 [\[4\]](#); Gray & Stroka, 2017 [\[5\]](#); Griffith et al., 2020 [\[6\]](#)).

We used nifedipine to temporally reduce heart rate and blood flow. Nifedipine is a blocker of L-type voltage-dependent calcium channels (VDCC) (Quevedo et al., 1998 [\[7\]](#)). A previous study shows that ML218, a T-type VDCC-selective inhibitor, tends to increase VSMC differentiation (Ando et al., 2022 [\[8\]](#)). In addition to the difference in drug targets, we also noted that in the previous study, the increase in VSMC differentiation only occur on anterior metencephalic central arteries (AMCtAs) that are more than 40 μm away from the BCA; these AMCtAs are much smaller than CoW arteries and have different geometry (Ando et al., 2022 [\[8\]](#)). Although the most obvious effect of nifedipine in zebrafish embryos is heart rate and blood flow reduction (Gierten et al., 2020 [\[9\]](#)), it is possible that nifedipine affect VSMC through VDCCs. Further investigation is needed to uncouple the roles of different VDCCs in VSMC differentiation and their association with hemodynamics.

We used *Tg(flt4:yfp, kdrl:ras-mcherry)^{hu4881/s896}* to show CoW arterial specification from primitive venous ECs (Chi et al., 2008 [\[10\]](#); Hogan et al., 2009 [\[11\]](#)). *kdrl* is not the best arterial marker as its expression is only enriched but not restricted to arteries. While the use of the two fluorescence transgenic lines helped us establish the temporal sequence of CoW morphogenesis, arterial specification and VSMC differentiation, analysis of additional arterial and venous markers is needed to fully characterize arterial specification during vertebrate brain vascular development.

In summary, our work identifies flow induced endothelial *klf2a* activation as a mechanism that regulates spatiotemporal dynamics of VSMC differentiation on CoW arteries in the vertebrate brain. Our data may help advance approaches to regenerate differentiated VSMCs in cerebrovascular diseases. It would be important to further investigate mechanisms upstream and downstream of *klf2a* activation in brain arterial ECs and additional mechanotransduction pathways that may contribute to VSMC differentiation on brain arteries.

Material and Methods

Zebrafish husbandry and transgenic lines

Zebrafish were raised and maintained at 28.5 °C using standard methods. Protocols are approved by the Yale Institutional Animal Care and Use Committee (2020-11473). *TgBAC(pdgfrb:egfp)^{ncv22}* was generated by Ando et al., has been reported that the *pdgfrb:egfp*⁺ cells express other pericyte enriched marker abcc9 (ATP binding cassette subfamily C member 9) (Ando et al., 2016 [\[12\]](#); Ando et al., 2019 [\[13\]](#)). By this transgenic line, they also observed that *pdgfrb* mutant lack brain pericytes and exhibit loss of vascular smooth muscle coverage (Ando et al., 2021 [\[14\]](#)). Thus, we used *TgBAC(pdgfrb:egfp)^{ncv22}* to study mural cell coverage of vascular endothelium in zebrafish CoW arteries. All transgenic lines in **Table 1** [\[15\]](#) were established previously.

Transgenic Line	ID	Lab of Origin	Reference
Tg(<i>flt4:ypf</i>) ^{hu4881}	ZDB-ALT-100208-1	S. Schulte-Merker	Hogan et al., 2009
Tg(<i>kdrl:hras-mcherry</i>) ^{s896} [Tg(<i>kdrl:ras-mcherry</i>) ^{s896}]	ZDB-ALT-081212-4	D.Y.R. Stainier	Chi et al., 2008
Tg(<i>acta2:mcherry</i>) ^{ca8}	ZDB-ALT-120508-2	S.J. Childs	Whitesell et al., 2014
TgBAC(<i>pdgfrb:egfp</i>) ^{ncv22}	ZDB-ALT-160609-1	N. Mochizuki	Ando et al., 2016
Tg(<i>kdrl:cerulean</i>) ^{sd24}	ZDB-ALT-131024-2	D. Traver	Page et al., 2013
Tg(<i>kdrl:grcfp</i>) ^{zn1} [Tg(<i>kdrl:gfp</i>) ^{zn1}]	ZDB-ALT-051114-10	Zygogen Research Department	Cross et al., 2003
Tg(<i>klf2a:h2b-egfp</i>) ^{ig11}	ZDB-ALT-161017-10	J. Vermot	Steed et al., 2016
Tg(<i>fli1:negfp</i>) ^{y7} [Tg(<i>fli1:nls-gfp</i>) ^{y7}]	ZDB-ALT-060821-4	B.M. Weinstein	Roman et al., 2002
Tg(<i>gata1:dsRed</i>) ^{sd2}	ZDB-ALT-051223-6	L.I. Zon	Traver et al., 2003

Table 1.

List of zebrafish fluorescent transgenic lines used in the study

Confocal fluorescence microscopy

Zebrafish embryos were raised in 0.003% 1-phenyl-2-thiourea (PTU, phenylthiocarbamide, or n-phenylthiourea, Sigma P7629) from gastrulation stage to prevent pigmentation. Embryos imaged live by confocal fluorescence microscopy were anesthetized in 0.1% tricaine methanesulfonate (TMS, MS-222, or Syncline, Western Chemical, NC0872873) and mounted in 1% low melt agarose within glass bottom microwell dishes. Fluorescence images were captured with an upright Zeiss LSM 980 confocal microscope using a 20X objective.

Image analysis

Confocal fluorescence images were analyzed with Imaris microscopy image analysis software (Bitplane, Oxford Instruments). Average fluorescence intensity of mcherry in *Tg(flt4:yfp, kdrl:hrras-mcherry)^{hu4881/s896}* was estimated by generating volume objects covering the artery of interest with Surface module. Vessel lengths of the arteries were manually traced with Filament module. Numbers of *pdgfrb*⁺ mural cell progenitors, *acta2*⁺ vascular smooth muscle cells, and *klf2a*⁺ and *fli1*⁺ endothelial nucleus were counted with Spots module.

Morpholino injections

Morpholino antisense oligonucleotides were synthesized by Gene Tools. Morpholinos in [Table 2](#) were validated previously. Optimized dose injected into each embryo are listed. Uninjected siblings were used as controls.

Nifedipine treatment

Nifedipine was dissolved into 20 mM in dimethyl sulfoxide (DMSO). The stock solution was diluted into 25 μ M in egg water with 0.003% PTU to treat zebrafish embryos from 54 hpf to 3 dpf or 4 dpf. The stock solution was diluted into 20 μ M in egg water with 0.003% PTU to treat zebrafish embryos from 4 dpf to 5 dpf. The same volume of DMSO was added into egg water with 0.003% PTU for sibling control embryos.

Microangiography

Embryos were anesthetized in 0.1% tricaine methanesulfonate, placed on an agarose mold, and injected pericardially with 4 nL Tetramethylrhodamine Dextran (2,000,000 molecular weight, Thermo Fisher) at a concentration of 10 mg/mL. Subsequently, the vasculature of embryos was checked for Dextran fluorescence signal under a stereomicroscope (Olympus, MVX10). Embryos with Dextran fluorescence were mounted in 1% low melt agarose within glass bottom microwell dishes and imaged with confocal microscope.

Red blood cell velocity

Red blood cell (RBC) velocity was measured by used axial line scanning to analyze zebrafish in the *Tg(kdrl:gfp;gata1:DsRed)^{zn1/sd2}* background, in which blood vessels and RBCs are labeled with green and red fluorescent protein, respectively ([Cross et al., 2003](#); [Traver et al., 2003](#)). As described previously ([Barak et al., 2021](#)), this axial line scan was conducted in a Zeiss LSM980 confocal microscope. A line was drawn inside the lumen of CoW vessels and parallel to the longitudinal axis of the vessel. The average length of the line was 21.21 μ m. The scan speed was chosen as maximum per second for 20,000 cycles (total scan duration was 5.86 s; line scan time is 292.80 μ s). A kymograph was exported in TIFF format after every measurement by ZEN 3.5 and analyzed RBC recognition and segmentation by using Fiji (ImageJ). Firstly, the RBC kymography image was processed using ‘noise despeckle’ and ‘enhance contrast’ steps. The brightness and contrast were appropriately adjusted if needed. Next, the RBCs were segmented, thresholding was

Table 2.

List of morpholino antisense oligonucleotides used in the study

Morpholino	Sequence	ID	Reference	Dose (ng/embryo)
MO1-tnt2a	5' - CATGTTTGCTCTGATCTGAC ACGCA - 3'	ZDB-MRPHLNO-060317-4	Sehnert et al., 2002	0.35
MO1-klf2a	5' - GGACCTGTCCAGTTCATCC TTCCAC - 3'	ZDB-MRPHLNO-100610-8	Nicoli et al., 2010	11

adjusted, and the slope of the RBCs was calculated automatically using the ‘Analyze particle’ option. Finally, the velocity of each RBC was calculated using the following equation: velocity = $(1/\tan(\alpha)) \times (\text{length of the y axis of the kymograph}/\text{duration of the total scan})$. We averaged systolic peak velocities and end diastolic velocities to measure the mean RBC velocity in the CoW vessels.

Computational fluid dynamic (CFD) simulation

Computational fluid dynamic (CFD) simulation was performed according to the geometry model of the circle of Willis (CoW) at 54 hpf, 3 and 4 dpf as previously described with modification (ANSYS, 2014; Barak et al., 2021). First, three-dimensional (3D) geometry models of CoW arteries were reconstructed from confocal z-stack microangiography images of Dextran injected Tg(*kdr1:gfp*) embryos with the Filament module in Imaris (9.9.0, Bitplane, Oxford instruments). These CoW 3D geometry models were pre-processed in ANSYS SpaceClaim 2022 R1 software. For example, the computational mesh on the CoW vascular wall was created and the boundary conditions were set up for the simulation. Meshing orthogonal quality was calculated, and objects less than 0.01 were excluded. Inlets and outlets were determined in the 3D model for the flow simulation. Next, the pre-processed models were imported in ANSYS Fluent 2022 R1 software. The non-Newtonian behavior of blood flow was modeled using the Carreau model which was provided by ANSYS. Blood density was considered as 1,060 kg/m³. The computational simulation used average RBC velocity among the CoW arteries at 54 hpf (532.72 μm/s), 3 dpf (602.34 μm/s) and 4 dpf (552.83 μm/s), respectively. After the CoW geometry and numerical solution of blood flow were ready, the flow simulation was implemented with 200 iterations to identify the magnitude and distribution of the WSS in the CoW.

Cell Culture

Pooled primary Human Umbilical Vein Endothelial Cells (HUVEC) (PCS-100-010™, ATCC) were seeded at an initial concentration of 5,000 cells per cm², in 1x M119 media (110343-023, Gibco) supplemented with 16% FBS (Gibco), 84 μg/mL of heparin sodium salt (H3393, Sigma-Aldrich), 25 μg/mL of endothelial cell growth supplement (02-102, EMD Millipore Corp.) and 1x Antibiotic-antimycotic solution (15240-062, Gibco). Cell cultures were maintained at 37°C, 5% CO₂ and 95% humidity, until the cell reached 80% confluence.

To facilitate cell visualization in cell co-cultures, Human Brain Vascular Pericytes (HBVP) (1200, ScienCell) were transfected using lentiviral particles (LVP310, GenTarget, Inc.) to induce GFP expression under EF1a promoter. Cells cultures were initiated by seeding 5,000 cells per cm² in 175 cm² plastic flasks pre-coated with gelatin and 1x DMEM (11995-065, Gibco) supplemented with 10% FBS (Gibco), and 1x Antibiotic-antimycotic solution under the same cell culture conditions described above. When cell cultures were at 80% confluence, 200 μL containing 2 x 10⁶ GFP-lentiviral particles were added to each 175 flasks. After 72 hours, fresh cell culture media supplemented with 10 μg/mL of Blasticidin (15205, Sigma-Aldrich) were added to each flask to select the positive transfected cells.

Flow assays and immunostaining

HBVP cells were harvested and seeded at a concentration of 1.3 x 10⁵ cells in 0.4 optical plastic flow microslides (80176, Ibidi) precoated with 1 mg/mL gelatin and incubated for 24 hours under standard culture conditions. After the initial incubation, 100 μg/mL of collagen I (354249, Corning) diluted in DMEM cell culture media was added to the slides to create a thin layer on top of the HBVP cells. After 2 hours, the media was removed and 2.5 x 10⁵ HUVECs were seeded on top of the collagen I layer and incubated for additional 24 hours. After cell co-cultures were established, the slides were exposed to laminar (15 dyn/cm²) or pulsatile (12-15 dyn/cm²) flow for 24 hours, implementing a peristaltic pump adapted to produce different types of flow (Abello, Raghavan, Yien, & Stratman, 2022). After 24 hours, cultures were rinsed with 1x PBS and fixed for 30 minutes in 4% paraformaldehyde at room temperature. Cell cultures were immunostained with α-

Smooth muscle actin D4K9N XP[®] rabbit monoclonal antibody (19245, Cell Signaling), followed by Alexa Fluor 633 goat anti-rabbit IgG (A21071, Invitrogen). Confocal images were obtained using a 40x objective with a W1 Spinning Disk confocal microscope, a Fusion camera, and the Nikon Eclipse Ti2-E base. Fiji image processing software was used for image analysis and fluorescence intensity quantification.

Statistics

All statistical analyses were performed with GraphPad Prism (version 10.0.3). Mann–Whitney test was used to compare two groups to test mean differences (protein level, morpholino and nifedipine treatment). Two-way ANOVA followed by Tukey's multiple comparisons was used to test mean differences compared more than two groups under different conditions (average fluorescence intensity, cell number, vascular diameter). Ordinary one-way ANOVA with Tukey's multiple comparisons was used to test mean differences among three groups (RBC velocity, WSS, and cell number). Pearson correlation analysis was used to compare the strength and direction of a linear relationship between two variables with a covariance (WSS vs acta2+ EC, acta2+ EC vs klf2+ EC).

Acknowledgements

We thank the labs of S. Schulte-Merker, D.Y.R. Stainier, S.J. Childs, N. Mochizuki, D. Traver, J. Vermot, B.M. Weinstein, and L.I. Zon for sharing zebrafish transgenic lines. We thank N. Semanchik for all the assistance with zebrafish adult colonies and husbandry. Experiments in the manuscript were supported by R01NS109160 and R01DK118728 awarded to S.N. and AHA23POST1025829 post-doctoral fellowship awarded to I.F.X. The paper is based on a dissertation submitted by S.C. to fulfill in part the requirements for the degree of Doctor of Philosophy, Yale University. S.C. was supported by a Gruber Science Fellowship from Yale Graduate School of Arts and Sciences.

Author Contributions

S.C. and I.F.X. contributed to the experiments, the data analysis, and composed the figures. R.W. performed dextran microangiography experiments. J.A. and A.N.S. performed the *in vitro* experiments. S.C., I.F.X. and S.N. wrote the manuscript and prepared figures. J.A. and A.N.S. edited the paper.

Supplementary Figures and Supplementary Figure Legends

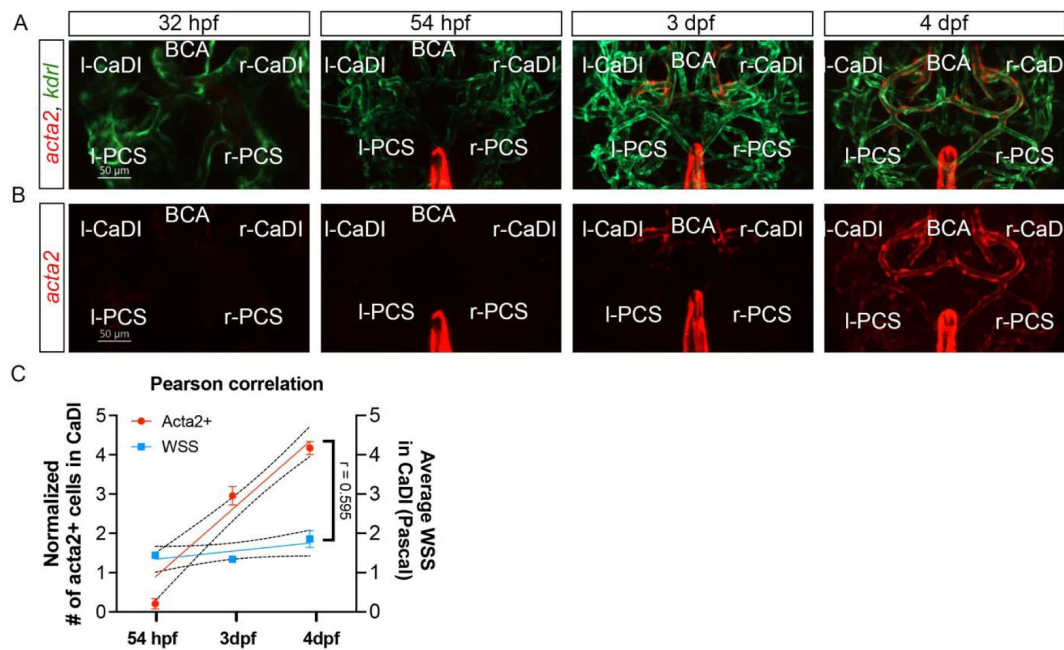


Figure S1.

Vascular smooth muscle cell (VSMC) differentiation on circle of Willis (CoW) arteries

(A-B) Confocal live images of Tg(*acta2:mcherry, kdrl:gfp*)^{ca8/zn1} in CoW arteries of zebrafish brain at 32 hour post fertilization (hpf), 54 hpf, 3 day post fertilization (dpf), and 4 dpf. Red channel represents *acta2:mcherry*, Green channel represents *kdrl:gfp*, and Merge panel combines both channels. Scale bar = 50 μ m

(C) Linear regression with normalized # *acta2*+ cells in CaDI (left y-axis) and average wall shear stress (WSS) (right y-axis) with the developmental stages (54 hpf, 3 and 4 dpf, x-axis), and Pearson correlation coefficient (r) is provided, indicating the degree of linear relationship between two variables.

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, VSMC: vascular smooth muscle cell, I-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, I-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

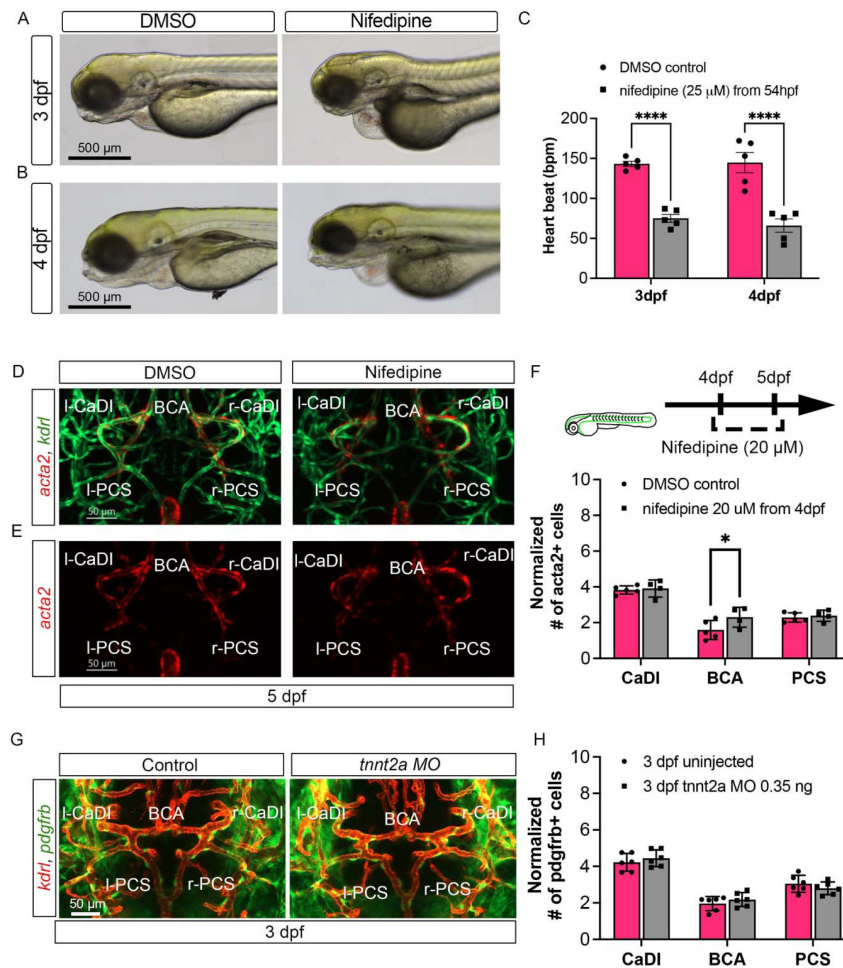


Figure S2.

Blood flow is not required for vascular smooth muscle cell (VSMC) maintenance or *pdgfrb*⁺ mural cell progenitor recruitment on circle of Willis (CoW) arteries

(A-B) Bright-field images of *Tg(acta2:mcherry; kdr1:gfp)^{ca8/zn1}* in control embryos treated with DMSO and embryos treated with 25 μM nifedipine from 54 hour post fertilization (hpf) at 3 day post fertilization (dpf) (A) and 4 dpf (B). Scale bar = 500 μm
(C) Quantitative heartbeat rate at 3 dpf and 4 dpf in DMSO control embryos and embryos treated with 25 μM nifedipine from 54 hpf (n=5 each group, 1 independent experiment), two-tailed Mann-Whitney test on each vessel's comparison, represented with mean ± SD, **** p≤0.0001

(D-E) Confocal live images of *Tg(acta2:mcherry; kdr1:gfp)^{ca8/zn1}* in CoW arteries of zebrafish brain at 5 dpf in DMSO control embryos and embryos treated with 20 μM nifedipine from 4 dpf. Red channel represents *acta2:mcherry*, Green channel represents *kdr1:gfp*, and Merge panel combines both channels. Scale bar = 50 μm

(F) Number of *acta2*⁺ VSMCs per 100 μm vessel length on caudal division of internal carotid arteries (CaDI), basal communicating artery (BCA), and posterior communicating segments (PCS) at 5 dpf in DMSO control (n=5, 1 independent experiment) and embryos treated with 20 μM nifedipine from 4 dpf (n=4, 1 independent experiment), two-tailed Mann-Whitney test on each vessel's comparison, represented with mean ± SD, * p≤0.05

(G) Confocal live images of *Tg(pdgfrb:egfp, kdr1:ras-mcherry)^{ncv22/s896}* labeling vascular endothelium and *pdgfrb*⁺ vascular mural cell progenitors on the CoW arteries in zebrafish brain at 3 dpf. Red channel represents *kdr1:ras-mcherry*, Green channel represents *pdgfrb:egfp*. Scale bar = 50 μm

(H) Number of *pdgfrb*⁺ vascular mural cell progenitors per 100 μm vessel length on CaDI, BCA, and PCS in control embryos and embryos injected with 0.35 ng *tnnt2a* morpholino (MO) at 3 dpf (n=6, 1 independent experiment), two-way analyses of variance followed by Tukey's multiple comparisons, represented with mean ± SD.

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, VSMC: vascular smooth muscle cell, I-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, I-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

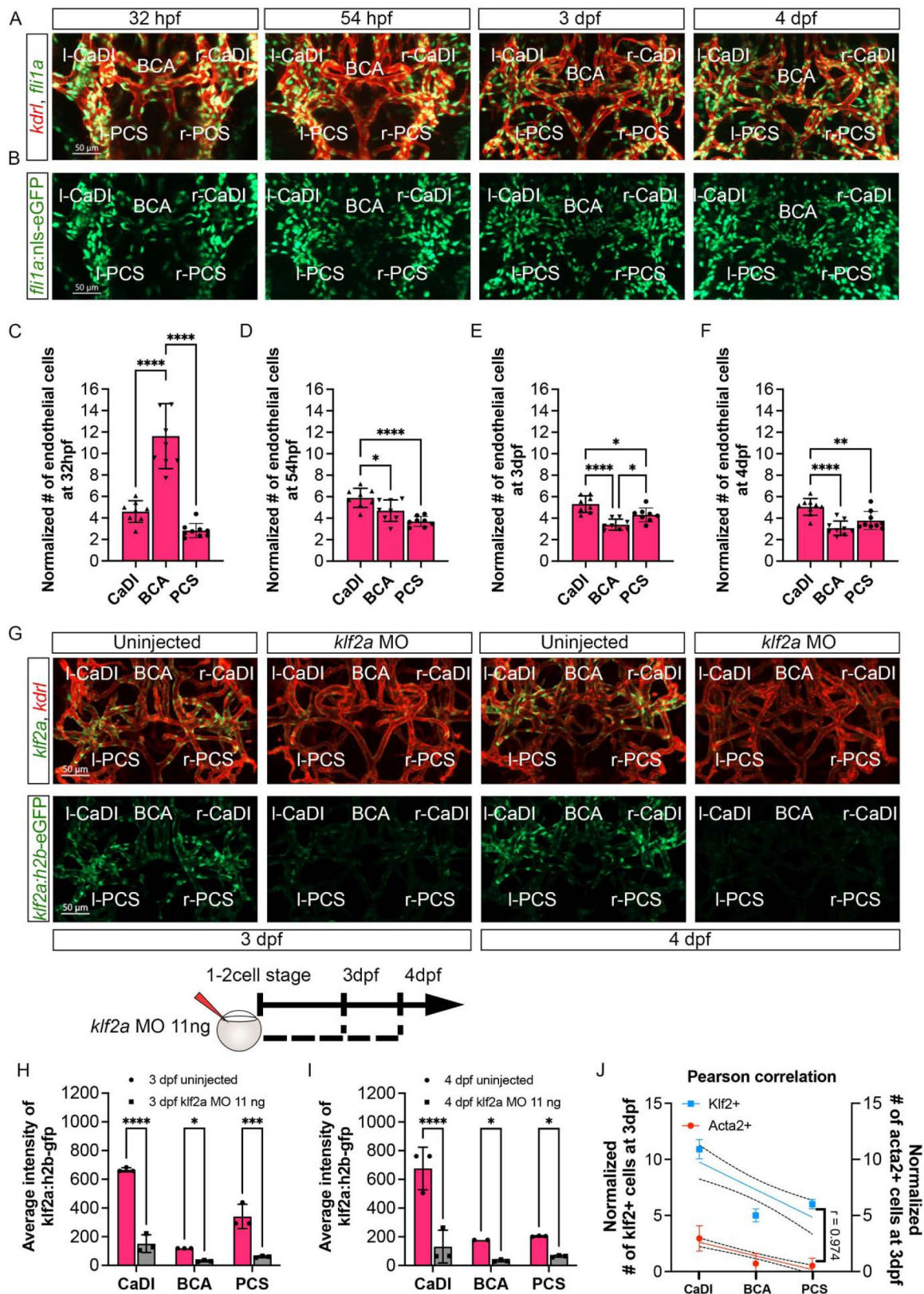


Figure S3.

Number of endothelial cells (ECs) in circle of Willis (CoW) arteries does not increase during *klf2a* activation

(A-B) Confocal live images of Tg(*fli1:nls-gfp, kdrl:ras-mcherry*)^{y7/s896} in CoW arteries of zebrafish brain at 32 hour post fertilization (hpf), 54 hpf, 3 day post fertilization (dpf), and 4 dpf. Red channel represents *kdrl:ras-mcherry* and Green channel represents *fli1:nls-gfp*. Scale bar = 50 μ m

(C-F) Number of ECs per 100 μ m vessel length on caudal division of internal carotid arteries (CaDI), basal communicating artery (BCA), and posterior communicating segments (PCS) at 32 hpf (n=9, 1 independent experiment) (C), 54 hpf (n=8, 1 independent experiment) (D), 3 dpf (n=8, 1 independent experiment) (E), and 4 dpf (n=9, 1 independent experiment) (F), ordinary one-way ANOVA analyses with Tukey's multiple comparisons, represented with mean $\pm$ SD, * $p \leq 0.05$, *** $p \leq 0.001$, **** $p \leq 0.0001$

(G) Confocal live images of Tg(*klf2a:h2b-gfp, kdrl:ras-mcherry*)^{ig11/s896} in CoW arteries of zebrafish brain at 3 and 4 dpf in uninjected control and embryos injected with 11 ng of *klf2a* morpholino (MO). Red channel represents *kdrl:ras-mcherry* and Green channel represents *klf2a:h2b-gfp*. Scale bar = 50 μ m

(H) Average intensity of *klf2a:h2b-gfp* in CaDI, BCA, and PCS at 3 dpf in uninjected control (n=3, 1 independent experiment) and embryos injected with 11 ng of *klf2a* MO (n=3, 1 independent experiment), two-tailed Mann-Whitney test on each vessel's comparison, represented with mean $\pm$ SD, * $p \leq 0.05$, *** $p \leq 0.001$, **** $p \leq 0.0001$

(I) Average intensity of *klf2a:h2b-gfp* in CaDI, BCA, and PCS at 4 dpf in uninjected control (n=3, 1 independent experiment) and embryos injected with 11 ng of *klf2a* MO (n=3, 1 independent experiment), two-tailed Mann-Whitney test on each vessel's comparison, represented with mean $\pm$ SD, * $p \leq 0.05$, **** $p \leq 0.0001$

(J) Linear regression with normalized # *klf2a*⁺ cells at 3 dpf (left y-axis) and normalized # *acta2*⁺ cells (right y-axis) with the different CoW arteries (CaDI, BCA and PCS, x-axis), and Pearson correlation coefficient (*r*) is provided, indicating the degree of linear relationship between two variables.

Abbreviations: hpf: hour post fertilization, dpf: day post fertilization, l-CaDI: left caudal division of internal carotid artery, r-CaDI: right caudal division of internal carotid artery, BCA: basal communicating artery, l-PCS: left posterior communicating segment, r-PCS: right posterior communicating segment

References

- Abello J., Raghavan S., Yien Y. Y., Stratman A. N. (2022) **Peristaltic pumps adapted for laminar flow experiments enhance in vitro modeling of vascular cell behavior** *J Biol Chem* **298** <https://doi.org/10.1016/j.jbc.2022.102404>
- Aguilar-Pineda J. A., Vera-Lopez K. J., Shrivastava P., Chavez-Fumagalli M. A., Nieto-Montesinos R., Alvarez-Fernandez K. L., Lino Cardenas C. L. (2021) **Vascular smooth muscle cell dysfunction contribute to neuroinflammation and Tau hyperphosphorylation in Alzheimer disease** *iScience* **24** <https://doi.org/10.1016/j.isci.2021.102993>
- Albinsson S., Nordstrom I., Hellstrand P. (2004) **Stretch of the vascular wall induces smooth muscle differentiation by promoting actin polymerization** *J Biol Chem* **279**:34849–34855 <https://doi.org/10.1074/jbc.M403370200>
- Ando K., Fukuhara S., Izumi N., Nakajima H., Fukui H., Kelsh R. N., Mochizuki N. (2016) **Clarification of mural cell coverage of vascular endothelial cells by live imaging of zebrafish** *Development* **143**:1328–1339 <https://doi.org/10.1242/dev.132654>
- Ando K., Shih Y. H., Ebarasi L., Grosse A., Portman D., Chiba A., Lawson N. D. (2021) **Conserved and context-dependent roles for pdgfrb signaling during zebrafish vascular mural cell development** *Dev Biol* **479**:11–22 <https://doi.org/10.1016/j.ydbio.2021.06.010>
- Ando K., Tong L., Peng D., Vazquez-Liebanas E., Chiyoda H., He L., Betsholtz C. (2022) **KCNJ8/ABCC9-containing K-ATP channel modulates brain vascular smooth muscle development and neurovascular coupling** *Dev Cell* **57**:1383–1399 <https://doi.org/10.1016/j.devcel.2022.04.019>
- Ando K., Wang W., Peng D., Chiba A., Lagendijk A. K., Barske L., Betsholtz C. (2019) **Peri-arterial specification of vascular mural cells from naive mesenchyme requires Notch signaling** *Development* **146** <https://doi.org/10.1242/dev.165589>
- ANSYS (2014) **ANSYS. (2014). ANSYS Innovation Courses: 3D bifurcating artery.** Retrieved from <https://courses.ansys.com/index.php/courses/fluent-3d-bifurcating-artery/>
- Bahrami N., Childs S. J. (2020) **Development of vascular regulation in the zebrafish embryo** *Development* **147** <https://doi.org/10.1242/dev.183061>
- Barak T., Ristori E., Ercan-Sencicek A. G., Miyagishima D. F., Nelson-Williams C., Dong W., Gunel M. (2021) **PP1L4 is essential for brain angiogenesis and implicated in intracranial aneurysms in humans** *Nat Med* **27**:2165–2175 <https://doi.org/10.1038/s41591-021-01572-7>
- Basatemur G. L., Jorgensen H. F., Clarke M. C. H., Bennett M. R., Mallat Z. (2019) **Vascular smooth muscle cells in atherosclerosis** *Nat Rev Cardiol* **16**:727–744 <https://doi.org/10.1038/s41569-019-0227-9>
- Campbell B. C. V., De Silva D. A., Macleod M. R., Coutts S. B., Schwamm L. H., Davis S. M., Donnan G. A. (2019) **Ischaemic stroke** *Nat Rev Dis Primers* **5** <https://doi.org/10.1038/s41572-019-0118-8>

- Cao W., Zhang D., Li Q., Liu Y., Jing S., Cui J., Yu B. (2017) **Biomechanical Stretch Induces Inflammation, Proliferation, and Migration by Activating NFAT5 in Arterial Smooth Muscle Cells** *Inflammation* **40**:2129–2136 <https://doi.org/10.1007/s10753-017-0653-y>
- Chandra S. M., Razavi H., Kim J., Agrawal R., Kundu R. K., de Jesus Perez V., Chun H. J. (2011) **Disruption of the apelin-APJ system worsens hypoxia-induced pulmonary hypertension** *Arterioscler Thromb Vasc Biol* **31**:814–820 <https://doi.org/10.1161/ATVBAHA.110.219980>
- Chen X., Gays D., Milia C., Santoro M. M. (2017) **Cilia Control Vascular Mural Cell Recruitment in Vertebrates** *Cell Rep* **18**:1033–1047 <https://doi.org/10.1016/j.celrep.2016.12.044>
- Chen Z., Qin H., Liu J., Wu B., Cheng Z., Jiang Y., Wang Y. (2019) **Characteristics of Wall Shear Stress and Pressure of Intracranial Atherosclerosis Analyzed by a Computational Fluid Dynamics Model: A Pilot Study** *Front Neurol* **10** <https://doi.org/10.3389/fneur.2019.01372>
- Chi N. C., Shaw R. M., De Val S., Kang G., Jan L. Y., Black B. L., Stainier D. Y. (2008) **Foxn4 directly regulates tbx2b expression and atrioventricular canal formation** *Genes Dev* **22**:734–739 <https://doi.org/10.1101/gad.1629408>
- Chou E. L., Lino Cardenas C. L., Chaffin M., Arduini A. D., Juric D., Stone J. R., Lindsay M. E. (2022) **Vascular smooth muscle cell phenotype switching in carotid atherosclerosis** *JVS Vasc Sci* **3**:41–47 <https://doi.org/10.1016/j.jvssci.2021.11.002>
- Colijn S., Nambara M., Stratman A. N. (2023) **Identification of overlapping and distinct mural cell populations during early embryonic development** *bioRxiv* <https://doi.org/10.1101/2023.04.03.535476>
- Cross L. M., Cook M. A., Lin S., Chen J. N., Rubinstein A. L. (2003) **Rapid analysis of angiogenesis drugs in a live fluorescent zebrafish assay** *Arterioscler Thromb Vasc Biol* **23**:911–912 <https://doi.org/10.1161/01.ATV.0000068685.72914.7E>
- Crouch E. E., Bhaduri A., Andrews M. G., Cebrian-Silla A., Diafos L. N., Birrueta J. O., Huang E. J. (2022) **Ensembles of endothelial and mural cells promote angiogenesis in prenatal human brain** *Cell* **185**:3753–3769 <https://doi.org/10.1016/j.cell.2022.09.004>
- Dekker R. J., van Soest S., Fontijn R. D., Salamanca S., de Groot P. G., VanBavel E., Horrevoets A. J. (2002) **Prolonged fluid shear stress induces a distinct set of endothelial cell genes, most specifically lung Kruppel-like factor (KLF2)** *Blood* **100**:1689–1698 <https://doi.org/10.1182/blood-2002-01-0046>
- Donadon M., Santoro M. M. (2021) **The origin and mechanisms of smooth muscle cell development in vertebrates** *Development* **148** <https://doi.org/10.1242/dev.197384>
- Duchemin A. L., Vignes H., Vermot J. (2019) **Mechanically activated piezo channels modulate outflow tract valve development through the Yap1 and Klf2-Notch signaling axis** *Elife* **8** <https://doi.org/10.7554/eLife.44706>
- Fernandes M. C., Sousa L. C., de Castro C. F., da Palma J. M. L. M., António C. C., Pinto S. I. S. (2022) **Implementation and Comparison of Non-Newtonian Viscosity Models in Hemodynamic Simulations of Patient Coronary Arteries** *Theoretical Analyses, Computations, and Experiments of Multiscale Materials: A Tribute to Francesco dell'Isola* :403–428

- Fontana F., Haack T., Reichenbach M., Knaus P., Puceat M., Abdelilah-Seyfried S. (2020) **Antagonistic Activities of Vegfr3/Flt4 and Notch1b Fine-tune Mechanosensitive Signaling during Zebrafish Cardiac Valvulogenesis** *Cell Rep* **32** <https://doi.org/10.1016/j.celrep.2020.107883>
- Fox B. M., Dorschel K. B., Lawton M. T., Wanebo J. E. (2021) **Pathophysiology of Vascular Stenosis and Remodeling in Moyamoya Disease** *Front Neurol* **12** <https://doi.org/10.3389/fneur.2021.661578>
- Frosen J., Joutel A. (2018) **Smooth muscle cells of intracranial vessels: from development to disease** *Cardiovasc Res* **114**:501–512 <https://doi.org/10.1093/cvr/cvy002>
- Fujita M., Cha Y. R., Pham V. N., Sakurai A., Roman B. L., Gutkind J. S., Weinstein B. M. (2011) **Assembly and patterning of the vascular network of the vertebrate hindbrain** *Development* **138**:1705–1715 <https://doi.org/10.1242/dev.058776>
- Gierten J., Pylatiuk C., Hammouda O. T., Schock C., Stegmaier J., Wittbrodt J., Loosli F. (2020) **Automated high-throughput heartbeat quantification in medaka and zebrafish embryos under physiological conditions** *Sci Rep* **10** <https://doi.org/10.1038/s41598-020-58563-w>
- Goddard L. M., Duchemin A. L., Ramalingan H., Wu B., Chen M., Bamezai S., Kahn M. L. (2017) **Hemodynamic Forces Sculpt Developing Heart Valves through a KLF2-WNT9B Paracrine Signaling Axis** *Dev Cell* **43**:274–289 <https://doi.org/10.1016/j.devcel.2017.09.023>
- Goldschmidt M. E., McLeod K. J., Taylor W. R. (2001) **Integrin-mediated mechanotransduction in vascular smooth muscle cells: frequency and force response characteristics** *Circ Res* **88**:674–680 <https://doi.org/10.1161/hh0701.089749>
- Gray K. M., Stroka K. M. (2017) **Vascular endothelial cell mechanosensing: New insights gained from biomimetic microfluidic models** *Semin Cell Dev Biol* **71**:106–117 <https://doi.org/10.1016/j.semcdb.2017.06.002>
- Griffith C. M., Huang S. A., Cho C., Khare T. M., Rich M., Lee G. H., Polacheck W. J. (2020) **Microfluidics for the study of mechanotransduction** *J Phys D Appl Phys* **53** <https://doi.org/10.1088/1361-6463/ab78d4>
- Haga J. H., Li Y. S., Chien S. (2007) **Molecular basis of the effects of mechanical stretch on vascular smooth muscle cells** *J Biomech* **40**:947–960 <https://doi.org/10.1016/j.jbiomech.2006.04.011>
- Heckel E., Boselli F., Roth S., Krudewig A., Belting H. G., Charvin G., Vermot J. (2015) **Oscillatory Flow Modulates Mechanosensitive klf2a Expression through trpv4 and trpp2 during Heart Valve Development** *Curr Biol* **25**:1354–1361 <https://doi.org/10.1016/j.cub.2015.03.038>
- Hill R. A., Tong L., Yuan P., Murikinati S., Gupta S., Grutzendler J. (2015) **Regional Blood Flow in the Normal and Ischemic Brain Is Controlled by Arteriolar Smooth Muscle Cell Contractility and Not by Capillary Pericytes** *Neuron* **87**:95–110 <https://doi.org/10.1016/j.neuron.2015.06.001>
- Hogan B. M., Herpers R., Witte M., Helotera H., Alitalo K., Duckers H. J., Schulte-Merker S. (2009) **Vegfc/Flt4 signalling is suppressed by Dll4 in developing zebrafish intersegmental arteries** *Development* **136**:4001–4009 <https://doi.org/10.1242/dev.039990>

- Isogai S., Horiguchi M., Weinstein B. M. (2001) **The vascular anatomy of the developing zebrafish: an atlas of embryonic and early larval development** *Dev Biol* **230**:278–301 <https://doi.org/10.1006/dbio.2000.9995>
- Kaplan L., Chow B. W., Gu C. (2020) **Neuronal regulation of the blood-brain barrier and neurovascular coupling** *Nat Rev Neurosci* **21**:416–432 <https://doi.org/10.1038/s41583-020-0322-2>
- Kathuria S., Gregg L., Chen J., Gandhi D. (2011) **Normal cerebral arterial development and variations** *Semin Ultrasound CT MR* **32**:242–251 <https://doi.org/10.1053/j.sult.2011.02.002>
- Katrtsis D., Kaiktsis L., Chaniotis A., Pantos J., Efstathopoulos E. P., Marmarelis V. (2007) **Wall shear stress: theoretical considerations and methods of measurement** *Prog Cardiovasc Dis* **49**:307–329 <https://doi.org/10.1016/j.pcad.2006.11.001>
- Lee J. S., Yu Q., Shin J. T., Sebzda E., Bertozzi C., Chen M., Kahn M. L. (2006) **Klf2 is an essential regulator of vascular hemodynamic forces in vivo** *Dev Cell* **11**:845–857 <https://doi.org/10.1016/j.devcel.2006.09.006>
- Leung D. Y., Glagov S., Mathews M. B. (1976) **Cyclic stretching stimulates synthesis of matrix components by arterial smooth muscle cells in vitro** *Science* **191**:475–477 <https://doi.org/10.1126/science.128820>
- Menshawi K., Mohr J. P., Gutierrez J. (2015) **A Functional Perspective on the Embryology and Anatomy of the Cerebral Blood Supply** *J Stroke* **17**:144–158 <https://doi.org/10.5853/jos.2015.17.2.144>
- Milewicz D. M., Kwartler C. S., Papke C. L., Regalado E. S., Cao J., Reid A. J. (2010) **Genetic variants promoting smooth muscle cell proliferation can result in diffuse and diverse vascular diseases: evidence for a hyperplastic vasculomyopathy** *Genet Med* **12**:196–203 <https://doi.org/10.1097/GIM.0b013e3181cdd687>
- Milewicz D. M., Trybus K. M., Guo D. C., Sweeney H. L., Regalado E., Kamm K., Stull J. T. (2017) **Altered Smooth Muscle Cell Force Generation as a Driver of Thoracic Aortic Aneurysms and Dissections** *Arterioscler Thromb Vasc Biol* **37**:26–34 <https://doi.org/10.1161/ATVBAHA.116.303229>
- Mirabella T., MacArthur J. W., Cheng D., Ozaki C. K., Woo Y. J., Yang M., Chen C. S. (2017) **3D-printed vascular networks direct therapeutic angiogenesis in ischaemia** *Nat Biomed Eng* **1** <https://doi.org/10.1038/s41551-017-0083>
- Nicoli S., Standley C., Walker P., Hurlstone A., Fogarty K. E., Lawson N. D. (2010) **MicroRNA-mediated integration of haemodynamics and Vegf signalling during angiogenesis** *Nature* **464**:1196–1200 <https://doi.org/10.1038/nature08889>
- Nowotarski S. H., Sánchez Alvarado A. (2016) **Widening perspectives on regenerative processes through growth** . *npj Regenerative Medicine* **1** <https://doi.org/10.1038/npjregenmed.2016.15>
- Oka M., Shimo S., Ohno N., Imai H., Abekura Y., Koseki H., Aoki T. (2020) **Dedifferentiation of smooth muscle cells in intracranial aneurysms and its potential contribution to the pathogenesis** *Sci Rep* **10** <https://doi.org/10.1038/s41598-020-65361-x>

- Page D. M., Wittamer V., Bertrand J. Y., Lewis K. L., Pratt D. N., Delgado N., Traver D. (2013) **An evolutionarily conserved program of B-cell development and activation in zebrafish** *Blood* **122**:e1–11 <https://doi.org/10.1182/blood-2012-12-471029>
- Parmar K. M., Larman H. B., Dai G., Zhang Y., Wang E. T., Moorthy S. N., Garcia-Cardena G. (2006) **Integration of flow-dependent endothelial phenotypes by Kruppel-like factor 2** *Clin Invest* **116**:49–58 <https://doi.org/10.1172/JCI24787>
- Poittevin M., Lozeron P., Hilal R., Levy B. I., Merkulova-Rainon T., Kubis N. (2014) **Smooth muscle cell phenotypic switching in stroke** *Transl Stroke Res* **5**:377–384 <https://doi.org/10.1007/s12975-013-0306-x>
- Quevedo J., Vianna M., Daroit D., Born A. G., Kuyven C. R., Roesler R., Quillfeldt J. A. (1998) **L-type voltage-dependent calcium channel blocker nifedipine enhances memory retention when infused into the hippocampus** *Neurobiol Learn Mem* **69**:320–325 <https://doi.org/10.1006/nlme.1998.3822>
- Rasouli S. J., El-Brolosy M., Tsedeke A. T., Bensimon-Brito A., Ghanbari P., Maischein H. M., Stainier D. Y. (2018) **The flow responsive transcription factor Klf2 is required for myocardial wall integrity by modulating Fgf signaling** *Elife* **7** <https://doi.org/10.7554/eLife.38889>
- Robin Y. M., Penel N., Perot G., Neuville A., Velasco V., Ranchere-Vince D., Coindre J. M. (2013) **Transgelin is a novel marker of smooth muscle differentiation that improves diagnostic accuracy of leiomyosarcomas: a comparative immunohistochemical reappraisal of myogenic markers in 900 soft tissue tumors** *Mod Pathol* **26**:502–510 <https://doi.org/10.1038/modpathol.2012.192>
- Roman B. L., Pham V. N., Lawson N. D., Kulik M., Childs S., Lekven A. C., Weinstein B. M. (2002) **Disruption of acvrl1 increases endothelial cell number in zebrafish cranial vessels** *Development* **129**:3009–3019 <https://doi.org/10.1242/dev.129.12.3009>
- Schröder H., Moser N., Huggenberger S. (2020) **The Mouse Circle of Willis** *Neuroanatomy of the Mouse: An Introduction* :333–340
- Sehnert A. J., Huq A., Weinstein B. M., Walker C., Fishman M., Stainier D. Y. (2002) **Cardiac troponin T is essential in sarcomere assembly and cardiac contractility** *Nat Genet* **31**:106–110 <https://doi.org/10.1038/ng875>
- Sharif-Naeini R., Folgering J. H., Bichet D., Duprat F., Lauritzen I., Arhatte M., Honore E. (2009) **Polycystin-1 and -2 dosage regulates pressure sensing** *Cell* **139**:587–596 <https://doi.org/10.1016/j.cell.2009.08.045>
- Steed E., Faggianelli N., Roth S., Ramspacher C., Concordet J. P., Vermot J. (2016) **klf2a couples mechanotransduction and zebrafish valve morphogenesis through fibronectin synthesis** *Nat Commun* **7** <https://doi.org/10.1038/ncomms11646>
- Stratman A. N., Burns M. C., Farrelly O. M., Davis A. E., Li W., Pham V. N., Weinstein B. M. (2020) **Chemokine mediated signalling within arteries promotes vascular smooth muscle cell recruitment** *Commun Biol* **3** <https://doi.org/10.1038/s42003-020-01462-7>
- Stratman A. N., Pezoa S. A., Farrelly O. M., Castranova D., Dye L. E., Butler M. G., Weinstein B. M. (2017) **Interactions between mural cells and endothelial cells stabilize the developing zebrafish dorsal aorta** *Development* **144**:115–127 <https://doi.org/10.1242/dev.143131>

- Stratman A. N., Weinstein B. M. (2021) **Assessment of Vascular Patterning in the Zebrafish** *Methods Mol Biol* **2206**:205–222 https://doi.org/10.1007/978-1-0716-0916-3_15
- Traver D., Paw B. H., Poss K. D., Penberthy W. T., Lin S., Zon L. I. (2003) **Transplantation and in vivo imaging of multilineage engraftment in zebrafish bloodless mutants** *Nat Immunol* **4**:1238–1246 <https://doi.org/10.1038/ni1007>
- Turczynska K. M., Hellstrand P., Sward K., Albinsson S. (2013) **Regulation of vascular smooth muscle mechanotransduction by microRNAs and L-type calcium channels** *Commun Integr Biol* **6** <https://doi.org/10.4161/cib.22278>
- Ulrich F., Ma L. H., Baker R. G., Torres-Vazquez J. (2011) **Neurovascular development in the embryonic zebrafish hindbrain** *Dev Biol* **357**:134–151 <https://doi.org/10.1016/j.ydbio.2011.06.037>
- Vila Cuenca M., Cochrane A., van den Hil F. E., de Vries A. A. F., Lesnik Oberstein S. A. J., Mummery C. L., Orlova V. V. (2021) **Engineered 3D vessel-on-chip using hiPSC-derived endothelial- and vascular smooth muscle cells** *Stem Cell Reports* **16**:2159–2168 <https://doi.org/10.1016/j.stemcr.2021.08.003>
- Wang Y., Cao W., Cui J., Yu Y., Zhao Y., Shi J., Liu J. (2018) **Arterial Wall Stress Induces Phenotypic Switching of Arterial Smooth Muscle Cells in Vascular Remodeling by Activating the YAP/TAZ Signaling Pathway** *Cell Physiol Biochem* **51**:842–853 <https://doi.org/10.1159/000495376>
- Watterston C., Zeng L., Onabadejo A., Childs S. J. (2019) **MicroRNA26 attenuates vascular smooth muscle maturation via endothelial BMP signalling** *PLoS Genet* **15** <https://doi.org/10.1371/journal.pgen.1008163>
- Whitesell T. R., Chrystal P. W., Ryu J. R., Munsie N., Grosse A., French C. R., Childs S. J. (2019) **foxc1 is required for embryonic head vascular smooth muscle differentiation in zebrafish** *Dev Biol* **453**:34–47 <https://doi.org/10.1016/j.ydbio.2019.06.005>
- Whitesell T. R., Kennedy R. M., Carter A. D., Rollins E. L., Georgijevic S., Santoro M. M., Childs S. J. (2014) **An alpha-smooth muscle actin (acta2/alphasma) zebrafish transgenic line marking vascular mural cells and visceral smooth muscle cells** *PLoS One* **9** <https://doi.org/10.1371/journal.pone.0090590>
- Wilson E., Sudhir K., Ives H. E. (1995) **Mechanical strain of rat vascular smooth muscle cells is sensed by specific extracellular matrix/integrin interactions** *J Clin Invest* **96**:2364–2372 <https://doi.org/10.1172/JCI118293>
- Wu J., Bohanan C. S., Neumann J. C., Lingrel J. B. (2008) **KLF2 transcription factor modulates blood vessel maturation through smooth muscle cell migration** *J Biol Chem* **283**:3942–3950 <https://doi.org/10.1074/jbc.M707882200>
- Zhong H., Wu X., Huang H., Fan Q., Zhu Z., Lin S. (2006) **Vertebrate MAX-1 is required for vascular patterning in zebrafish** *Proc Natl Acad Sci U S A* **103**:16800–16805 <https://doi.org/10.1073/pnas.0603959103>

Editors

Reviewing Editor

Sarah Childs

University of Calgary, Canada

Senior Editor

Didier Stainier

Max Planck Institute for Heart and Lung Research, Bad Nauheim, Germany

Reviewer #2 (Public Review):

Summary:

Cheng et al. explore the development of the arteries that form the circle of Willis and investigate how blood flow pulsatility influences vascular smooth muscle cell (VSMC) differentiation. Using live confocal imaging of the developing zebrafish, the authors show that endothelial cells in circle of Willis arteries transition from venous to arterial identity between 54 hours post-fertilization (hpf) and 3 days post-fertilization (dpf), and that this coincides with *pdgfrb*⁺ mural cell progenitor differentiation into *acta2*⁺ arterial VSMCs. They find that the anterior portions of the circle of Willis, including the internal carotid arteries (CaDI), establish *acta2* expression earlier than posterior aspects, likely due to faster flow rate and increased pulsatility through the CaDI. Then, using computational fluid dynamics, an in vitro co-culture assay, and genetic and drug manipulations of blood flow, the authors provide evidence that *pdgfrb*⁺ differentiation is dependent upon pulsatile blood flow and *klf2a* activation. The results add to our understanding of vascular development and suggest that deficits in pulsatile flow could be potential drivers of arteriopathies.

Strengths:

- (1) Longitudinal confocal imaging of live developing zebrafish makes the timeline of arterial development in the circle of Willis easy to understand. This is a strong approach to studying how vascular networks are altered with genetic and pharmacological manipulations.
- (2) Rigorous use of multiple techniques to test the hypothesis that pulsatile blood flow is required for smooth muscle cell differentiation. The microangiography experiment, in vitro co-culture assay, and genetic and drug manipulations of heart rate at various developmental timepoints yield outcomes that are consistent with the hypothesis.

Weaknesses:

- (1) The authors should provide more information on how blood flow velocity and wall shear stress are calculated from circle of Willis vascular structure. It is presumed that these values are dependent upon the 3-D morphology of the vessel network, as labeled by intravenous dextran dye, but this is not clear. Small local differences in vessel diameter and shape will influence blood flow velocity, but these morphological changes are not clearly articulated. Further, it is unclear how flow input levels to the CaDI and basilar arteries are decided across time-points. In general, descriptions of the blood flow modeling are very sparse.
- (2) Is it possible to measure the blood flow speed empirically with line-scanning or high-speed tracking of labeled blood cells? This would provide some validation of the modeling results.
- (3) Does the cardiac injection of dextran itself affect the diameter or flow of the arteries, given the invasiveness of the procedure? This could be examined in fish with a transgenic endothelial label and with vs. without dextran.
- (4) The data from the microangiography experiment in Figure 3 does not fully support the stated results. The authors report that the CaDI had the highest blood flow speed starting from 54 hpf, but it does not appear to be higher than the other arteries at this time point.

Additionally, there is not sufficient evidence that wall shear stress coincides with smooth muscle cell differentiation in the CaDI. Wall shear stress appears to be similar between 54 hpf and 3 dpf in the CaDI, only increasing between 3 dpf and 4 dpf, while differentiation is shown to begin at 3 dpf.

(5) The genetic and drug manipulations of heart rate are important experiments, but more detail is required to understand the effects of the manipulations. At least, a discussion on the limitations of these manipulations is needed. For example, how does one separate the pulsatile versus nutritive effects of blood flow/heart rate reduction? It is possible that off-target or indirect effects of Nifedipine decrease smooth muscle cell proliferation, or that altered cardiac contractility fundamentally alters many aspects of vascular development other than blood flow. Nifedipine is also likely to act upon VSMC calcium handling in the circle of Willis, which may in turn affect cell maturation.

(6) It is unclear if *acta2* expression is conferring vascular tone, as would be expected if the cells are behaving as mature VSMCs. Does arterial diameter decrease with an increase in *acta2* expression? Are *acta2* positive mural cells associated with more dynamic changes in arteriole diameter under basal or stimulated conditions?

<https://doi.org/10.7554/eLife.94094.2.sa1>

Reviewer #3 (Public Review):

Summary:

Cheng et al. studied if and how blood flow regulates differentiation of vascular smooth muscle cells (VSMC) in the Circle of Willis (CW) in zebrafish embryos. They show that CW vessels gradually acquire arterial identity. VSMCs also undergo gradual differentiation, which correlates with blood flow velocity. Using cell culture they show that pulsatile blood flow promotes pericyte differentiation into smooth muscle cells. They further identify transcription factor *klf2a* as differentially regulated by blood flow, and show that *klf2a* inhibition results in VSMC differentiation. The authors conclude that pulsatile flow promotes VSMC differentiation through *klf2a* activation.

Strengths:

Overall this is an important study, because VSMC differentiation in CW has not been previously studied, although analogous observations regarding the role of blood flow and *klf2* involvement have been previously made in other systems and other vascular beds, for example, mouse *klf2* mutants, which have deficient VSMC coverage of the dorsal aorta (Wu et al., 2008, JBC 283: 3942-50). The results convincingly show that VSMC differentiation in CW depends on the blood flow, and that *klf2a* flow dependent function regulates VSMC differentiation.

Weaknesses:

- (1) The provided data do not support correlation between wall shear stress (WSS) and *acta2*⁺ cell number. The number of *acta2*⁺ cells in CaDI increases dramatically between 54 hpf and 3 dpf (Fig. 2F). However, the graph provided in the response to reviewers shows that WSS in CaDI is actually lower at 3 dpf compared to 54 hpf. Authors argue that Pearson correlation analysis shows that both variables increase together, but this is calculated over the stage between 54 hpf and 4 dpf. *acta2*⁺ cells appear by 3 dpf, and at this stage WSS in CaDI is not increased (or even lower), which argues against WSS being the cause of *acta2*⁺ cell differentiation. Furthermore, data in Fig. 3I-K show that WSS actually decreases in BCA and PCS between 54 hpf and 4 dpf, while the number of *acta2*⁺ increases in BCA and PCS by 4 dpf. This also argues against the argument that WSS affects differentiation of *acta2*⁺ cells.
- (2) In multiple instances, results are based on a single independent experiment (Fig. 3, Fig.

4H, I, Fig. S2 and Fig. S3) with only a few embryos analyzed in many cases. This falls short of expected standards in the field, and it is unclear if these results are reproducible.

<https://doi.org/10.7554/eLife.94094.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Weaknesses to be addressed:

(1) More detail is required to understand the effects of genetic and drug manipulations on heart rate as these are important experiments. At the very least, a discussion on the limitations of these manipulations is needed.

- For example, how does one separate the pulsatile versus nutritive effects of blood flow/heart rate reduction?

- The conclusion that arterial SMC differentiation is driven by pulsatile blood flow needs to be toned down. Indeed, this conclusion is mainly supported by in vitro cell co-cultures exposed to laminar versus pulsatile flow. In vivo, reducing Tnnt2a expression affects cardiac contractility and blood flow does not selectively affect pulsatility. To make this conclusion, the authors would need an experimental means to selectively dampen the pulsatility of blood flow.

We understand this concern and we toned down the statements related to the pulsatile flow of our conclusion by using 'flow' instead of 'pulsatile flow' in all text except for the in vitro co-cultures part. We also added a paragraph to discuss the limited capability of qualitatively reduce blood flow in vivo, and acknowledge that the effects of nutrients and flow reduction could not be uncoupled in live zebrafish embryos. We proposed that in the future, in vitro 3D vascular culture models may be combined with microfluidics to precisely calibrate nutrient composition in culture media, flow velocity and pulse; these methods would help address these questions more thoroughly. See page 11-12 line 312-322.

(2) Since mural cells are sensitive to transmural pressure, could the authors elaborate on the potential role of raised intravascular pressure in SMC differentiation? This would better parallel rodents and humans.

We thank you for this suggestion. We added a paragraph to discuss the potential role of raised intravascular pressure in VSMC differentiation in the discussion section (see page 11 line 296-311).

(3) The authors use nifedipine to reduce blood flow. Nifedipine is a specific and potent inhibitor of voltage-dependent calcium channels (VDCC) which are expressed in SMCs. Prior studies (PMID: 35588738) showed that VDCC blockers increased rather than inhibited SMC differentiation. Nifedipine is also likely to act upon VSMC calcium handling in the circle of Willis, which may in turn affect cell maturation. Could the authors comment on this seeming discrepancy?

It is possible that off-target or indirect effects of Nifedipine decrease smooth muscle cell proliferation, or that altered cardiac contractility fundamentally alters aspects of vascular development other than blood flow.

- Additionally, it would be helpful to report the quantitative heart rate reduction achieved with Nifedipine. This would clear up concerns that the heart rate reduction is too large

for normal vascular development to occur, and thus decrease proliferation rate independent of changes in blood flow pulsatility.

We concur with these comments, which is why our experimentation with Nifedipine is reinforced by employing an alternative, non-pharmacological strategy to inhibit blood flow: the use of morpholino against *tnnt2a* gene. The results with either Nifedipine or *tnnt2a* support the lack of VSMCs maturation. In addition, we provided the quantitative heart rate reduction achieved with Nifedipine shown in new Figure S2A-S2C, suggesting that the drug is not completely halting the heart rate but decreasing it. Nevertheless, we report that Zebrafish embryos can survive and develop a normal blood vascular system without any heartbeat. Hence, we exclude that the effect on VSMCs maturation is linked non-specific effects caused by the loss of heartbeat. Nevertheless, we now acknowledged in our discussion the limitation of nifedipine, as it may affect VSMC through VDCCs (page 12, line 323-334).

We also added a paragraph in the discussion section to compare nifedipine, an L-type VDCC blocker, and ML218, a T-type VDCC selective inhibitor from the previous study (Ando et al., 2022). We noted that in this previous study, the increase in VSMC differentiation only occur on anterior metencephalic central arteries (AMCtAs) that are more than 40 mm away from the BCA; these AMCtAs are much smaller than CoW arteries and have different geometry hence possible different kinetics of VSMC maturation (Ando et al., 2022) as our manuscript discovery would suggest.

(4) The authors should provide more information on how blood flow velocity and wall shear stress are calculated from the Circle of Willis vascular structure. It is presumed that these values are dependent upon the 3-D morphology of the vessel network, as labeled by intravenous dextran dye, but this is not clear. (a second reviewer similarly comments: I was unclear how flow velocity values were obtained in Fig. 3E. Are they based on computational simulation, or are they experimentally calculated following the dextran injection?) Small local differences in vessel diameter and shape will influence blood flow velocity, but these morphological changes are not clearly articulated. Further, it is unclear how flow input levels to the CaDI and basilar arteries are decided across time points. For instance, is it possible to measure the blood flow speed empirically with line-scanning or high-speed tracking of labeled blood cells or particles? This would provide validation of the modeling results.

The computational fluid dynamic simulation was performed according to previous study from our lab (Barak et al., 2021). Blood flow velocity and wall shear stress are dependent upon the 3D morphology of the vessel network labeled by intravascular dextran. Details on how the computational fluid dynamic simulation was performed are added in method section page 17 line 433-449.

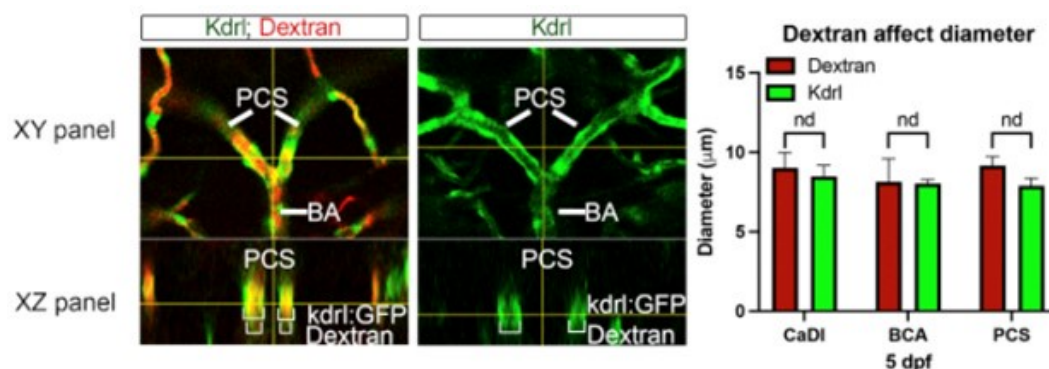
Moreover, to address this reviewer concern we have now provided new experimental measurement of blood flow using the red blood cell (RBC) velocity via axial line scanning microscopy in *Tg(kdrl:gfp;gata1:DsRed)zn1/sd2* zebrafish embryos at 54 hpf, 3 dpf, and 4 dpf. By using the experimental RBC velocity, we re-simulated the computational fluid dynamic. The new findings align with our conclusion and are further elaborated upon in response to this reviewer comment listed as point 6. Details on how RBC velocity calculated is added in method section page 16 line 414-431.

(5) Does the cardiac injection of dextran itself affect the diameter of the arteries, given the invasiveness of the procedure? This could be examined in fish with a transgenic endothelial label with and without dextran.

Here, we performed an experiment on wildtype zebrafish at 5 days post-fertilization (dpf) with and without Dextran injection, examining the effects of Dextran injection on vessel

diameters. As shown in the representative image below, the XZ panel clearly illustrates a Dextran-filled PCS vessel with no alteration in vessel size. Dextran microangiography, a technique employed to obtain vessel geometry with fluorescent microsphere, has been well established in zebrafish (Kamei et al., 2010). Our findings, demonstrating that Dextran does not affect vessel size, are consistent with previous studies utilizing Dextran microangiography.

Author response image 1.

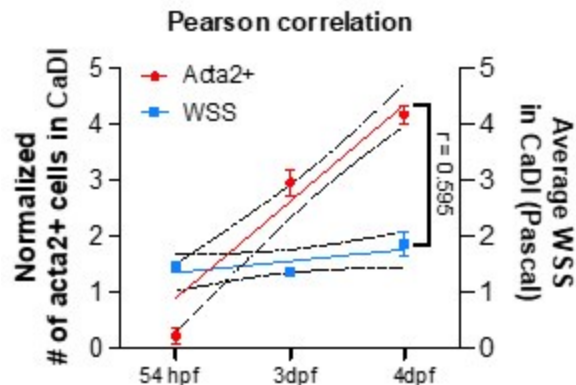


(6) The data from the microangiography experiment in Figure 3 does not fully support the stated results. The authors report that the CaDI had the highest blood flow speed starting from 54 hpf, but it does not appear to be higher than the other arteries at this time point. Additionally, there is not sufficient evidence that wall shear stress coincides with smooth muscle cell differentiation in the CaDI. Wall shear stress appears to be similar between 54 hpf and 3 dpf in the CaDI, only increasing between 3 dpf and 4 dpf, while differentiation is shown to begin at 3 dpf. The authors need to address this and/or soften conclusions.

First, In response to this specific reviewer concern, we measured red blood cell (RBC) velocity by used axial line scanning microscopy to analyze Tg(kdrl:gfp;gata1:DsRed)zn1/sd2 zebrafish embryos (the detailed method was added in Method section in the manuscript). We replaced the computational simulated blood flow velocity by RBC velocity in new Figure 3E-3G, and re-run the computational simulated wall shear stress (WSS) using the RBC velocity in new Figure 3I-3K. We compared RBC velocity and WSS among different vessels at each time point. We confirmed that CaDI has the highest RBC velocity starting from 54 hpf to 4 dpf (new Figure 3A-3C, and 3E-3G) and found an overall increase in average WSS from 54 hpf to 4 dpf (new Figure 3A-3C, and 3H). Further, WSS in CaDI was significantly higher than BCA and PCS at 54 hpf, 3 dpf, and 4 dpf (new Figure 3A-3C, 3I-3K). Altogether, the CFD simulation suggests that CoW arteries experience different hemodynamic WSS that is associated with spatiotemporal pattern of VSMC differentiation on CoW arteries.”. (Page 6, line 153-162)

Second, to identify the correlation of WSS and VSMC differentiation in CaDI, we performed Pearson correlation analysis. In the image provided here, we plotted a linear regression with normalized # of acta2+ cells in CaDI and WSS with developmental stages (54 hpf, 3 and 4 dpf), and performed Pearson correlation coefficient analysis by using GraphPad Prism 10.0.3. The correlation coefficient $r = 0.595$, suggesting that the two variables (acta2+ cells and WSS) tend to increase together with developmental stages (54 hpf, 3 and 4 dpf).

Author response image 2.



Third, we softened our conclusion as the RBC velocity across CoW arteries was differentially distributed while VSMC differentiation occurred in these vessels.

(7) It is unclear if *acta2* expression is conferring vascular tone, as would be expected if the cells are behaving as mature VSMCs. Does arterial diameter decrease with an increase in *acta2* expression? Are *acta2*-positive mural cells associated with more dynamic changes in arteriole diameter under basal or stimulated conditions?

Thanks for this interesting question. VSMC maturation and its vasoactivity could be further investigated in the future. Our study focused on early stage of VSMC differentiation, in which *pdgfrb*⁺ progenitors started to express VSMC marker *acta2*. We discussed the onset of transgelin expression and loss of *abcc9* expression as markers of VSMC maturation. In addition, a previous study found that VSMC covered vessels in zebrafish brain dilate as early as 4 dpf and constrict at 6 dpf (Bahrami & Childs, 2020). Future study may focus on the association between expression of different VSMC markers and VSMC functional maturation. (page 10, line 272-279)

(8) The authors argue that CoW vessels transition from venous to arterial identity (Fig. 1). However, *kdrl* is not an ideal arterial marker for this experiment as it is expressed in both arteries and veins. While it is true that many arterial beds have stronger *kdrl* expression than the veins, its expression in both arteries and veins changes with developmental stage, and its expression level may vary depending on the type of vessel. Therefore, showing that *kdrl* increases from 32 hpf - 4 dpf in CoW vessels is not convincing because its expression may increase in both venous or arterial vasculature as the vessels mature. In addition, *flt4* expression is not exclusively venous; for example, it has noticeable expression in the dorsal aorta at 24-32 hpf stages. It would be helpful to confirm this transition by analyzing additional arterial and venous markers.

We acknowledge this and we added a paragraph to discuss the limitation. We combined loss of *flt4* and increase in *kdrl* to establish the temporal sequence of circle of Willis morphogenesis, arterial specification, and VSMC differentiation. We acknowledge that additional arterial and venous markers need to be analyzed for a more thorough characterization of arterial specification in vertebrate brain vascular development. See page 12 line 335-341.

(9) The authors show that *acta2*⁺ VSMCs are absent in *tnnt2a* MO embryos, concluding that blood flow is required for their differentiation from pericytes. However, there is no

data showing that pericytes are still present in tnnt2a MO embryos. Although this has been previously shown by Ando et al 2016, it would be beneficial to confirm in the current study as this is a critical piece of evidence needed for this conclusion.

To determine if blood flow is dispensable for pdgfrb+ progenitor recruitment, we performed tnnt2a MO (0.35 ng/embryo) injection in Tg(pdgrb:egfp, kdrl:ras-mcherry) ncv22/s896. Loss of blood flow did not affect pdgfrb+ progenitor emergence around the CoW (new Figure S2G-S2H) at 3 days post fertilization (dpf). This is consistent with previous observation in Ando et al 2016 Figure S2C (Ando et al., 2016).

(10) The authors show that klf2a MO injected embryos have a reduced number of VSMCs at 3 dpf but a normal number at 4 dpf (Fig. 6), concluding that klf2a is only important to initiate CaDI muscularization. If this is true, it would raise important questions about how VSMCs differentiate at a later stage in the absence of klf2a. For instance, is blood flow not required to differentiate at a later stage, or is there another factor that compensates in the absence of klf2a? The alternative explanation/ caveat is that klf2a MO loses efficacy with development, leading to the recovery of VSMCs at this stage. Therefore, it would be important to confirm this result using a genetic klf2a mutant.

Thank you for pointing this out. We note that based on the klf2a reporter line, klf2a activity in CoW arterial endothelial cells is highly correlated with the number of acta2+ VSMCs in CaDI, BCA and PCS at 3 dpf ($r = 0.974$, new Figure S5J). Interestingly however, klf2a activity remained stable from 3 dpf to 4 dpf, well beyond initiation of VSMC differentiation. Thus, we speculate sustained klf2a expression may support further maturation of VSMCs, as acta2+ VSMCs showed distinct morphology at 4 dpf compared with 3 dpf. (Page 10, line 268-272). As for the observation that klf2a morphants have normal number of VSMCs at 4 dpf, we think that in addition to the temporary effect of morpholino, a proximal explanation is compensation by paralogous klf2b in zebrafish. We acknowledge that further characterization of CoW VSMC development in klf2a and klf2b double genetic mutants (Rasouli et al., 2018; Steed et al., 2016) may help determine whether klf2b compensates klf2a in CoW VSMC differentiation beyond 4 dpf. See page 10-11 line 292-295.

(11) A large part of the discussion focuses on Notch and Wnt signaling, as downstream Klf2 effectors. While these are reasonable hypotheses to propose, there is no data on the involvement of these pathways in the current study. It seems excessive to speculate on detailed mechanisms of how Klf2 activates Notch and Wnt signaling in the absence of data showing that these pathways are affected in CoW vessels. Therefore, the discussion could be shortened here unless additional data can be obtained to demonstrate the involvement of these pathways in VSMCs in CoW.

We concur and have condensed the discussion on Notch and Wnt signaling as downstream klf2 effectors.

Minor comments:

(1) Line 138 "CaDI is the only vessels in the CoW receiving pulsatile arterial blood flow ... ". Adding a reference to support this statement would be useful.

We agree and revised this sentence into 'CaDI receive proximal arterial feed through lateral dorsal aorta from cardiac outflow tract (Isogai et al., 2001)'. It was also based on our general observation of zebrafish vascular anatomy and blood flow under a confocal microscope.

(2) The image insets in Figs. 1A, 2A, 4E-L, 5A, 6A are quite small. Please make them larger to help the reader interpret the findings.

We agree. We maximized the image size to help the reader interpret the finding, and to visualize confocal images and schematics side-by-side.

(3) The schematics in Figs. 1-2, and 4-6 are helpful, but the different cell types are difficult to see because they are small and their colors/shapes are not very distinct.

We agree. We increased the size and color contrast to provide better visualization of the schematics in new schematic Figures. 1-2 and 4-6.

(4) It is stated that there are no diameter differences between different arteries, but statistics are not reported.

The statistics in Figure 3D were performed by ordinary two-way ANOVA followed by Tukey's multiple comparisons test, with a single pooled variance. Here we added pairwise comparisons among vessels in the CoW. Hence when non indicated the difference are non-significant.

(5) Figure 3F would be better visualized on a log scale, as it is difficult to see the differences between each post-fertilization timepoint.

We agree. In the new Figure 3H, the average wall shear stress (WSS) in CoW arteries is presented on log scale in y axis to see the differences between each post-fertilization timepoint.

(6) Please provide more background and validation on the pericyte cell line, and their use for the questions in this study.

Thank you for the question, TgBAC(pdgrb:egfp)ncv22 was generated and described by [Ando et al 2016](#) to clarify mural cell coverage of vascular endothelium in zebrafish (Ando et al., 2016). We added a describe in the method section to provide background and validation on this pericyte line (see page 13 line 368-372).

(7) Flow velocity and WSS changes are shown in each vessel in Figs. 3E,G. However, the comparison should be made between different types of vessels to see if there is a statistical difference and PCS, for example, which would explain differences in VSMC coverage.

We agreed. We compared the difference among arteries in the CoW at each developmental timepoint and performed ordinary one-way ANOVA with Tukey's multiple comparisons test. Figure. 3E is replaced by new Figure. 3E-G and Figure. 3G is replaced by new Figure. 3I-K.

(8) Similarly, between CaDI, the number of klf2a cells in Fig. 5B should be compared between different vessels, not between different stages of the same vessel.

We agree. In new Figure 5B-E, the number of klf2a+ cells per 100 μ m vessel length are compared among different vessels at each developmental stage and analyzed by ordinary one-way ANOVA with Tukey's multiple comparisons test.

(9) When quantifying klf2+ cells in Fig. 5, it would be helpful to quantify klf2 expression level between cells in different vessels. This could be done by quantifying GFP expression in existing images. The difference in expression level may explain the variation between CaDI and PCS more accurately than just the difference in cell number.

The GFP expression reflect the stability of GFP protein expression and labels discrete nuclei with active *klf2a* expression. Hence the quantification of GFP level might not give an accurate readout of *klf2a* expression per se but rather of its activity. For this reason we don't think that this experiment will add accurate measurement of *klf2a* expression.

(10) Do data points in Figure 4D correspond to different cells in the same chamber experiment? If so, they cannot be treated as independent replicates. Each data point should correspond to an independent replicate experiment.

We agree. Now in the figure legend, we report the number of cells analyzed.

*(11) Graph placement is confusing in Figs. 4I, M. An adjacent Fig. 4G shows Nifedipine treated embryos, while the graph next to (Fig. 4I) shows *acta+* cell number from *tnnt2a* 4 dpf experiment. Similarly, the bottom Fig. 4K *tnn2a* 4 dpf MO experiment has an adjacent graph Fig. 4M, which shows nifedipine treatment quantification, which makes it very confusing.*

We agreed. We rearranged Figure 4E (representative images of control embryos at 3 dpf and 4 dpf), Figure 4F (*tnnt2a* MO embryos at 3 dpf and 4 dpf), Figure 4G (nifedipine treated embryos at 3 dpf and 4 dpf).

Reference:

Ando, K., Fukuhara, S., Izumi, N., Nakajima, H., Fukui, H., Kelsh, R. N., & Mochizuki, N. (2016). Clarification of mural cell coverage of vascular endothelial cells by live imaging of zebrafish. *Development*, 143(8), 1328-1339. <https://doi.org/10.1242/dev.132654>

Ando, K., Tong, L., Peng, D., Vazquez-Liebanas, E., Chiyoda, H., He, L., Liu, J., Kawakami, K., Mochizuki, N., Fukuhara, S., Grutzendler, J., & Betsholtz, C. (2022). KCNJ8/ABCC9-containing K-ATP channel modulates brain vascular smooth muscle development and neurovascular coupling. *Dev Cell*, 57(11), 1383-1399 e1387. <https://doi.org/10.1016/j.devcel.2022.04.019>

Bahrami, N., & Childs, S. J. (2020). Development of vascular regulation in the zebrafish embryo. *Development*, 147(10). <https://doi.org/10.1242/dev.183061>

Barak, T., Ristori, E., Ercan-Sencicek, A. G., Miyagishima, D. F., Nelson-Williams, C., Dong, W., Jin, S. C., Prendergast, A., Armero, W., Henegariu, O., Erson-Omay, E. Z., Harman, A. S., Guy, M., Gultekin, B., Kilic, D., Rai, D. K., Goc, N., Aguilera, S. M., Gulez, B., . . . Gunel, M. (2021). PPIL4 is essential for brain angiogenesis and implicated in intracranial aneurysms in humans. *Nat Med*, 27(12), 2165-2175. <https://doi.org/10.1038/s41591-021-01572-7>

Isogai, S., Horiguchi, M., & Weinstein, B. M. (2001). The vascular anatomy of the developing zebrafish: an atlas of embryonic and early larval development. *Dev Biol*, 230(2), 278-301. <https://doi.org/10.1006/dbio.2000.9995>

Kamei, M., Isogai, S., Pan, W., & Weinstein, B. M. (2010). Imaging blood vessels in the zebrafish. In *Methods in cell biology* (Vol. 100, pp. 27-54). Elsevier.

Rasouli, S. J., El-Brolosy, M., Tsedeke, A. T., Bensimon-Brito, A., Ghanbari, P., Maischein, H. M., Kuenne, C., & Stainier, D. Y. (2018). The flow responsive transcription factor *Klf2* is required for myocardial wall integrity by modulating *Fgf* signaling. *Elife*, 7. <https://doi.org/10.7554/eLife.38889>

Steed, E., Faggianelli, N., Roth, S., Ramspacher, C., Concordet, J. P., & Vermot, J. (2016). *klf2a* couples mechanotransduction and zebrafish valve morphogenesis through fibronectin synthesis. *Nat Commun*, 7, 11646. <https://doi.org/10.1038/ncomms11646>

<https://doi.org/10.7554/eLife.94094.2.sa3>